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Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a meta-analysis

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Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a meta-analysis

Authors: Peng Yang[1,2#], Xiang-Dong Zheng[1,2#], Jun-Mei Wang[1,2], Wen-Bin Geng[1,2], Xiao-Yong Wang[1]*

1. Department of Gastroenterology, Changzhou NO.2 People's Hospital, Affiliated with Nanjing Medical University, 29 Xinglong lane, Tianning District, Changzhou 213000, Jiangsu Province, China.

2. Graduate school, Dalian Medical University, 9 West Section of Lushun South Road, Lvshunkou District, Dalian 116000, Liaoning Province, China.

[#]The authors contributed equally to this work.

*Corresponding author: Xiao-Yong Wang, MD, From the Department of Gastroenterology, Changzhou NO.2 People's Hospital, Affiliated with Nanjing Medical University, 29 Xinglong Lane, Tianning District, Changzhou 213000, Jiangsu Province, China. wxy20009@126.com

ORICD ID: https://orcid.org/0000-0001-8401-4888

Authors' ORICD ID:

Xiang-Dong Zheng : https://orcid.org/0000-0003-1088-6872

Peng Yang: https://orcid.org/0000-0002-5501-0244

Jun-Mei Wang: https://orcid.org/0000-0002-2117-4144

Wen-Bin Geng: https://orcid.org/0000-0001-6770-7690

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Abstract

Objectives: Previous studies have shown that differentiated-predominant mixed-type (MD) early gastric cancer (EGC) is more aggressive than pure differentiated-type (PD) EGC. However, the biological behavior of undifferentiated-predominant mixed-type (MU) EGC and pure undifferentiated-type (PU) EGC are controversial. The aim of this meta-analysis was to compare the biological behavior of MU EGC and PU EGC.

Design: Meta-analysis.

Methods: Literature searches were performed using PubMed and EmBase for relevant studies published from inception to June 2020. Relevant data were extracted, and the pooled results were expressed as odds ratios (ORs) with 95% confidence intervals (95% CIs) using Stata software.

Results: In total, 10 studies were included for analysis. Patients with MU EGC had significantly higher risks of lymph node metastasis (LNM) (OR, 2.61; 95% CI, 2.18–3.13), submucosal invasion (OR, 2.12; 95% CI, 1.77–2.54), and lymphovascular invasion (OR, 2.73; 95% CI, 2.12–3.50) compared with PU EGC patients. When the data were stratified according to region, significantly higher risks for LNM (OR, 3.04; 95% CI, 2.49-3.72) and lymphovascular invasion (OR, 2.94; 95% CI, 2.27–3.80) correlated to MU EGC were found in studies from Japan; and significantly higher risks for submucosal invasion (OR, 2.46; 95% CI, 1.70–3.55) correlated to MU EGC were found in studies from China. In addition, MU EGC exhibited significantly more depressed-type (IIc) endoscopic growth patterns (OR, 1.69; 95% CI, 1.10-2.58) compared with PU EGC.

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Conclusions: This study identified that patients with MU EGC had an increased risk of submucosal invasion, lymphovascular invasion, depressed-type (IIc) endoscopic growth patterns and LNM compared with PU EGC patients, which indicated that more attention should be focused on the clinical management of patients with MU EGC.

Keywords: lymph node metastasis; lymphovascular invasion; submucosal invasion; undifferentiated-type; mixed-type; early gastric cancer

Strengths and limitations of this study:

This is the first meta-analysis to compare the biological behavior between MU and PU EGC.

All included studies were carried out in Asia.

There might be some selection bias that cannot be completely excluded as all studies were retrospective.

The number of studies included in this meta-analysis was limited and the total number of patients included was relatively small.

Declarations

Acknowledgment:

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Background

Gastric cancer (GC) has become the fifth most common type of cancer and the third leading cause of death due to cancer in the world, and there were more than one million new cases of GC diagnosed in 2018[1]. The stage of GC at the time of diagnosis dictates patient prognosis; the five-year overall survival (OS) rate of early gastric cancer (EGC) is 92.0-97.2%[2], whereas for advanced GC, the five-year OS rate is only 23-36%[3]. Therefore, early detection, diagnosis and treatment of GC are essential to optimize patient survival. EGC is defined as cancer tissue limited to the mucosa and submucosa, regardless of lymph node metastasis (LNM)[4]. Endoscopic resection including endoscopic mucosal resection (EMR) and endoscopic submucosal dissection (ESD), as well as surgery are treatment options for EGC, whereas a gastrectomy with adequate lymphadenectomy and chemotherapy is the only therapeutic option for advanced GC[5]. ESD has become an important method of treatment for EGC patients with a lower risk of LNM[6,7]. It has been reported that the five-year OS and disease-specific survival (DSS) rates were not significantly different between patients that received ESD (OS: 96.4%, DSS: 99.6%) or surgery (OS: 97.2%, DSS: 99.2%)[2]. This was similar to another study that reported OS rates of 92.0% in patients who underwent ESD, and 90.2% in patients who underwent surgery[8]. Additionally, a meta-analysis involving 5993 patients revealed no significant differences in OS and DSS between EGC patients receiving ESD or surgery[9]. Importantly, these studies show that long-term outcomes are comparable between ESD and surgery, but ESD is known as a less invasive therapeutic method, has fewer postoperative complications, and improves patient quality of life[2,9]. Despite its distinct merits, the clinical success of ESD treatment depends on the true status of LNM.

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Consequently, assessing the possibility of LNM is a priority to decide whether patients should accept ESD treatment.

According to the Japanese Gastric Cancer Association[4], the histologic types of GC are classified as either differentiated-type or undifferentiated-type. The former consists of papillary adenocarcinoma and well-differentiated or moderately differentiated tubular adenocarcinoma, whereas the latter includes poorly differentiated adenocarcinoma, mucinous adenocarcinoma and signet-ring cell carcinoma. It has been reported that the undifferentiated histological type was associated with an increased risk of LNM[10,11]. Current indications of ESD for EGC are based on depth of invasion (intramucosal or submocosal), histological type (differentiated or undifferentiated), tumor size, and ulcer findings (yes or no)[7]. Absolute indication for ESD involves a differentiated-type GC without ulcer findings, in which the clinical diagnosis is intramucosal carcinoma and there is no limit on tumor size. In addition, if there are ulcer findings with a differentiated-type GC and the clinical diagnosis is intramucosal carcinoma with a tumor size ≤ 3 cm, the absolute indication is also applicable. When it comes to the expanded indication, if an undifferentiated-type GC is without ulcer findings and the clinical diagnosis is intramucosal carcinoma with a tumor size ≤ 2 cm, the expanded indication is adapted.

However, GC lesions usually show histological heterogeneity, and a mixture of different histological types are frequently observed in clinical practice. Approximately 12.3-20.7% of EGC patients have mixed-type[12-14], which consists of both differentiated and undifferentiated histological type according to the histology. Histological types can be divided into the following four groups according to the proportions of differentiated and

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undifferentiated components: pure differentiated-type (PD, differentiated components only), pure undifferentiated-type (PU, undifferentiated components only), differentiated-predominant mixed-type (MD, major differentiated and minor undifferentiated components), and undifferentiated-predominant mixed-type (MU, major undifferentiated and minor differentiated components). Previous studies have shown that MD EGC has a significantly higher risk of LNM and a higher rate of non-curative resection when compared to PD EGC[15-17]. However, the biological behavior of MU EGC and PU EGC is controversial. Certain reports have claimed that MU EGC has a significantly higher risk of LNM than PU EGC[18]. However, Chen *et al.* reported that MU EGC did not show an increase in the incidence of LNM compared to PU EGC[19]. Additionally, according to the treatment guidelines of the Japanese Gastric Cancer Association[7], MU and PU EGC are collectively treated as undifferentiated-type EGC. Differences in the biological behavior between PU and MU have not been given much consideration, and it remains unknown whether they should be treated using the same ESD indication criteria. It is necessary to combine data from the studies available to maximize the power of this research in a relatively rare circumstance to address this issue. The above-mentioned considerations prompted the generation of this meta-analysis to reveal the biological behaviors in MU and PU EGC and determine if there are important differences that can improve the clinical approach to treatment.

Methods

This meta-analysis was completed according to the principles of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-P)[20].

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Literature search

Studies from inception to June 2020 were searched using the PubMed and Embase databases. Due to language limitations, only studies published in English were selected. Two independent evaluators (Peng Yang and Xiang-dong Zheng) used common key words associated with "early gastric cancer" and "mixed type" to search the databases. The details of search strategy are shows in Supplementary_material. The titles and abstracts of each article were reviewed to exclude irrelevant publications, and full texts were read to determine whether the remaining studies meet the inclusion criteria. References of the included articles were carefully checked to ensure no literature was unintentionally omitted. Disagreements between the two evaluators were resolved by a third evaluator (Xiao-yong Wang). The detailed retrieval strategy is presented in **Fig.1**.

Eligibility criteria: Sufficient information was required from the studies to construct a 2 x 2 contingency table for further analysis, and available information was either extracted directly or calculated from the data contained by the studies. This meta-analysis included studies with the following criteria: (1) direct and indirect comparison of the relationship between PU and MU; (2) patients with EGC; (3) a specified outcome of LNM, lymphovascular invasion, submucosal invasion, and macroscopic aspect of the lesions and/or ulcer findings; and (4) the primary lesion was obtained by open surgery or laparoscopic surgery. For articles with overlapping data of the same source population, only the largest report was included.

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Data extraction

Pertinent information was extracted from each study independently by two reviewers (Peng Yang and Xiang-dong Zheng) who followed the same criteria. Results were compared, and disparities were settled by a third evaluator (Xiao-yong Wang). The essential study information included: name of the first author(s); date of publication; region from which the study population was derived. The necessary clinicopathologic factors included: LNM (yes or no); depth of tumor invasion (mucosal or submucosal invasion); lymphovascular invasion (yes or no); macroscopic aspect of the lesions (depressed-type [IIc] or non-depressed [other types]); and ulcer findings (yes or no).

Quality assessment

The quality of the included studies was assessed by two initial evaluators (Peng Yang and Xiang-dong Zheng) using the Newcastle-Ottawa scale (NOS). There major aspects including selection, comparability, and exposure/outcome were used to evaluate each paper. The final score ranged from 0 to 9, where 1 point was given for meeting one required item, and at most 2 points were given for comparability. Articles with a score ≥ 6 were considered high quality. Discrepancies were resolved by a third evaluator (Xiao-yong Wang).

Statistical analysis

Stata software, version 15.1 (StataCorp LLC 4905 Lakeway Drive, College Station, TX 77845 USA) was used to perform all statistical analyses.

The pooled rates were calculated for LNM, submucosal invasion, lymphovascular

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invasion, macroscopic aspect of the lesions, and unclear findings between MU and PU EGC groups, which were expressed as odds ratios (ORs) with 95% confidence intervals (95% CIs). In addition to the overall comparisons, a region-based subgroup analysis was also implemented.

Heterogeneity of the studies was assessed using both the I^2 test and Cochran's Q test. Heterogeneity was considered statistically significant if $P < 0.05$ or $I^2 > 50\%$. If $P > 0.05$ or $I^2 \leq 50\%$, a fixed-effects model was performed; otherwise, a random-effects model was used. Publication bias was tested using both Begg's and Egger's tests. In addition, a sensitivity analysis was performed to assess the stability and reliability of the results.

Patient and public involvement

Not applicable.

Results

Literature identification

As shown in Figure 1, a total of 282 publications were retrieved through the initial literature search. Among these search results, 63 duplicates were removed, which left 219 for screening based on title and abstract. Two hundred and ten were excluded for not meeting the inclusion criteria as stated in the methods section. Nine publications were assessed based on the full text, and two additional publications were included after reading the references of these nine publications in detail[21,22]. One publication was excluded due to a different definition of PU (only undifferentiated components *vs.* more than 90% undifferentiated components)[23]. After

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the thorough screening, 10 publications [13,18,19,21,22,24-28] were included for this meta-analysis, and the basic summarizing information from these articles is shown in **Table 1**.

Meta-analysis

The risk of LNM and subgroup analysis

Nine studies were used to compare PU and MU EGC with regard to LNM risk. There was no significant heterogeneity between the included studies (I^2 =44.3%; P-heterogeneity, 0.073); therefore, a fixed-effects model was used. Importantly, MU EGC was found to have an increased risk of LNM (OR, 2.61; 95% CI, 2.18–3.13) when compared to PU EGC (**Fig.2**). As shown in **Fig.3**, when considering only the seven studies with patients from Japan, a significantly higher risk for LNM was found to be associated with MU EGC (OR, 3.04; 95% CI, 2.49-3.72). There was no statistically significant heterogeneity between these seven studies (I^2 = 0.0%; P-heterogeneity, 0.960). Therefore, a fixed-effects model was performed.

The risk of submucosal invasion and subgroup analysis

Four studies were used to compare PU and MU EGC with respect to submucosal invasion risk, and a fixed-effects model for the meta-analysis was applied since there was no statistically significant heterogeneity (I^2 = 0.0%; P-heterogeneity, 0.794). MU EGC was found to have an increased risk of submucosal invasion (OR, 2.12; 95% CI, 1.77–2.54; **Fig.4**). As shown in **Fig.5**, MU EGC had a significantly higher submucosal invasion risk compared to PU EGC (OR, 2.46; 95% CI, 1.70–3.55) only in studies involving patients from China. A fixed-effects model was also applied for this analysis since there was no statistically

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significant heterogeneity between the included studies (I^2, 0.0%; P-heterogeneity, 0.828).

The risk of lymphovascular invasion and subgroup analysis

Four studies were used to compare PU and MU EGC with respect to lymphovascular invasion risk. Since there was no statistically significant heterogeneity between the four studies (I^2 =46.0%; P-heterogeneity, 0.135), a fixed-effects model was applied. MU EGC was associated with an increased risk of lymphovascular invasion (OR, 2.73; 95% CI, 2.12–3.50; **Fig.6**). As indicated in **Fig.7**, MU EGC had a significantly higher lymphovascular invasion risk compared to PU EGC (OR, 2.94; 95% CI, 2.27–3.80) only in studies involving patients from Japan.

The risk of depressed (IIc) endoscopic growth patterns

Three studies were used to compare PU and MU EGC with respect to macroscopic aspects of the lesions, and MU significantly exhibited more frequent depressed (IIc) endoscopic growth patterns (OR, 1.69; 95% CI, 1.10-2.58; **Fig.8**). There was no statistically significant heterogeneity between studies (I^2 = 0.0%; P-heterogeneity, 0.790).

Data regarding the ulcer findings was also extracted to compare MU and PU EGC, but no statistically significant difference was found.

Sensitivity analysis and publication bias

To assess publication bias of the included studies, Begg's and Egger's tests were performed. There was no evidence of publication bias in our meta-analysis for risk of LNM (Begg's test:

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P=0.917; Egger's test: P=0.910), submucosal invasion (Begg's test: P = 0.734; Egger's test: P = 0.680), lymphovascular invasion (Begg's test: P = 0.308; Egger's test: P = 0.138) or macroscopic aspect of the lesions (Begg's test: P = 1.000; Egger's test: P=0.139). Therefore, it can be suggested that the publication bias for the included studies was relatively small. Furthermore, the results of our meta-analysis may represent real differences in the biological behavior between MU and PU EGC.

A sensitivity analysis was also performed by removing one study at one time. The results concerning LNM, submucosal invasion risk, lymphovascular invasion, and macroscopic aspect of the lesions were not reversed with the sequential removal of each study. Therefore, it can be concluded that MU EGC is associated with a higher risk of LNM, submucosal invasion, lymphovascular invasion, and depressed (IIc) endoscopic growth patterns as compared to PU EGC.

Discussion

ESD has been gaining popularity as the primary treatment for patients with EGC. However, since ESD procedures do not include lymph node resection, indications for ESD are only applicable to EGC patients with negligible risk of LNM. Therefore, the ability to accurately assess the risk of LNM is crucial. It has been reported that mixed-type EGC has a significant correlation with LNM risk and contributes to malignant outcomes when compared to non-mixed-type EGC[29,30]. Previous studies have reported that MD EGC shows more aggressive biological behavior compared to PD EGC, but controversy and ambiguity remain as to whether MU and PU EGC have distinct biological behaviors. The aim of this study was

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to compile existing knowledge, gain insight, and provide clear guidance for the clinical management of MU and PU EGC. This meta-analysis confirmed that MU and PU EGC display different biological behaviors. Patients with MU EGC had a higher significantly risk of LNM, submucosal invasion, and lymphovascular invasion compared with PU EGC patients. Consequently, more attention should be given to patients with MU EGC for clinical management of the disease. Careful assessment is essential for patients with MU EGC before they undergo an ESD procedure.

The biological mechanism whereby MU is more aggressive than PU EGC is inconclusive, but previous studies can provide insight and propose areas of research to be investigated. From the viewpoint of molecular pathology, the genomic instability subtype of GC has been shown to be associated with different histological types: the chromosomal instability subtype of GC was significantly associated with increased histological numbers and mixed-type GC, whereas the genomic stability subtype of GC showed a significant relationship with pure-type GC[31]. Another study reported that the protein expression levels of Ki-67, extracellular matrix metalloproteinase inducer (EMMPRIN) and vascular endothelial growth factor (VEGF), which participate in angiogenesis and cell proliferation, were reduced in intestinal-type GC compared to the intestinal portion of mixed-type GC[32]. From the viewpoint of epigenetics, it has been reported that the elevated methylation levels of CpG islands, which inhibit the expression of tumor suppressor genes, may promote the development of mixed-type GC over diffuse-type or intestinal-type GC[33]. Taken together, genomic instability, the expression level of related genes, and elevated methylation levels of CpG islands might contribute to the more aggressive behavior of MU compared with PU

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EGC. Although the exact mechanisms remain unknown, additional articles exploring the mechanism of this aggressive biological behavior of MU EGC would provide valuable insight in the future.

Previous studies have reported the biological behaviors and clinicopathologic features that differ between PU and MU EGC, but controversies remain. This meta-analysis brings together all the available information to draw a more substantial conclusion. However, the present study has several limitations. Firstly, all included studies were carried out in Asia, with the majority from Japan. Consequently, extending these results beyond Asia may be inaccurate and require further study with data from different races and regions. Secondly, all studies were retrospective. Therefore, there might be some selection bias that cannot be completely excluded. Thirdly, the number of studies included in this meta-analysis was limited, and the total number of patients included was relatively small. As a result, the statistical effectiveness of the meta-analysis is somewhat low. Nevertheless, despite these limitations, this is the first meta-analysis to compare the biological behavior between MU and PU EGC. Additional large-scale, multicenter studies are expected to validate these results.

Conclusions

In conclusion, this study identified that patients with MU EGC had an increased risk of LNM, submucosal invasion, lymphovascular invasion, and depressed (IIc) endoscopic growth patterns compared with PU EGC patients. In future clinical practice, extra effort should be dedicated to further substantiate the outcomes of the present study. These results will help doctors to select the appropriate clinical management and therapy options for patients with MU EGC to improve their outcomes.

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REFERENCES

1. Bray F, Ferlay J, Soerjomataram I, et al.. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2018; 68: 394-424. 10.3322/caac.21492

2. Hahn KY, Park CH, Lee YK, et al.. Comparative study between endoscopic submucosal dissection and surgery in patients with early gastric cancer. *Surg Endosc*. 2018; 32: 73-86. 10.1007/s00464-017-5640-8

3. Cunningham D, Allum WH, Stenning SP, et al.. Perioperative chemotherapy versus surgery alone for resectable gastroesophageal cancer. *N Engl J Med*. 2006; 355: 11-20. 10.1056/NEJMoa055531

4. Japanese Gastric Cancer Association. Japanese classification of gastric carcinoma: 3rd English edition. *Gastric Cancer*. 2011; 14: 101-12. 10.1007/s10120-011-0041-5

5. Smyth EC, Nilsson M, Grabsch HI, et al.. Gastric cancer. *Lancet*. 2020; 396: 635-648. 10.1016/S0140-6736(20)31288-5

6. Takizawa K, Ono H, Muto M. Current indications of endoscopic submucosal dissection for early gastric cancer in Japan. *Jpn J Clin Oncol*. 2019; 49: 797-802. 10.1093/jjco/hyz100

7. Japanese Gastric Cancer Association. Japanese gastric cancer treatment guidelines 2018 (5th edition). *Gastric Cancer*. 2021; 24: 1-21. 10.1007/s10120-020-01042-y

8. Cho JH, Cha SW, Kim HG, et al.. Long-term outcomes of endoscopic submucosal dissection for early gastric cancer: a comparison study to surgery using propensity score-matched analysis. *Surg Endosc*. 2016; 30: 3762-73. 10.1007/s00464-015-4672-1

9. Liu Q, Ding L, Qiu X, et al.. Updated evaluation of endoscopic submucosal dissection

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versus surgery for early gastric cancer: A systematic review and meta-analysis. *Int J Surg*. 2020; 73: 28-41. 10.1016/j.ijsu.2019.11.027

10. Gotoda T, Yanagisawa A, Sasako M, et al.. Incidence of lymph node metastasis from early gastric cancer: estimation with a large number of cases at two large centers. *Gastric Cancer*. 2000; 3: 219-225. 10.1007/pl00011720

11. Chen L, Wang YH, Cheng YQ, et al.. Risk factors of lymph node metastasis in 1620 early gastric carcinoma radical resections in Jiangsu Province in China: A multicenter clinicopathological study. *J Dig Dis*. 2017; 18:556-565. 10.1111/1751-2980.12545

12. Takeuchi H, Abe N, Hashimoto Y, et al.. Establishment of pathological quantitative method for determining undifferentiated component ratio in patients with differentiated/undifferentiated mixed-type early gastric cancer and clinical significance of this ratio. *Gastric Cancer*. 2018; 21: 661-671. 10.1007/s10120-017-0782-x

13. Zhong Q, Sun Q, Xu GF, et al.. Differential analysis of lymph node metastasis in histological mixed-type early gastric carcinoma in the mucosa and submucosa. *World J Gastroenterol*. 2018; 24: 87-95. 10.3748/wjg.v24.i1.87

14. Zhao B, Huang R, Lu H, et al.. Risk of lymph node metastasis and prognostic outcome in early gastric cancer patients with mixed histologic type. *Curr Probl Cancer*. 2020; 44: 100579. 10.1016/j.currproblcancer.2020.100579

15. Min BH, Kim KM, Park CK, et al.. Outcomes of endoscopic submucosal dissection for differentiated-type early gastric cancer with histological heterogeneity. *Gastric Cancer*. 2015 ; 18: 618-26. 10.1007/s10120-014-0378-7

16. Shim CN, Chung H, Park JC, et al.. Early gastric cancer with mixed histology

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predominantly of differentiated type is a distinct subtype with different therapeutic outcomes of endoscopic resection. *Surg Endosc*. 2015; 29: 1787-94. 10.1007/s00464-014-3861-7

17. Horiuchi Y, Fujisaki J, Yamamoto N, et al.. Undifferentiated-type component mixed with differentiated-type early gastric cancer is a significant risk factor for endoscopic non-curative resection. *Dig Endosc.* 2018; 30: 624-632. 10.1111/den.13059

18. Horiuchi Y, Ida S, Yamamoto N, et al.. Feasibility of further expansion of the indications for endoscopic submucosal dissection in undifferentiated-type early gastric cancer. *Gastric Cancer*. 2020; 23: 285-292. 10.1007/s10120-019-01003-0

19. Chen JN, Wang QW, Zhang QW, et al.. Poorly differentiated is more significant than signet ring cell component for lymph node metastasis in mixed-type early gastric cancer: a retrospective study from a large-volume hospital. *Surg Endosc*. 2020. 10.1007/s00464-020-07532-5

20. Shamseer L, Moher D, Clarke M, et al.. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ*. 2015; 350: g7647. 10.1136/bmj.g7647

21. Hanaoka N, Tanabe S, Mikami T, et al.. Mixed-histologic-type submucosal invasive gastric cancer as a risk factor for lymph node metastasis: feasibility of endoscopic submucosal dissection. *Endoscopy*. 2009; 41: 427-32. 10.1055/s-0029-1214495

22. Miyamae M, Komatsu S, Ichikawa D, et al.. Histological mixed-type as an independent risk factor for nodal metastasis in submucosal gastric cancer. *Tumour Biol*. 2016; 37: 709-14. 10.1007/s13277-015-3864-6

23. Seo HS, Lee GE, Kang MG, et al.. Mixed Histology Is a Risk Factor for Lymph Node

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Metastasis in Early Gastric Cancer. *J Surg Res*. 2019; 236: 271-277. 10.1016/j.jss.2018.11.055

24. Ito H, Inoue H, Ikeda H, et al.. Clinicopathological characteristics and treatment strategies in early gastric cancer: a retrospective cohort study. *J Exp Clin Cancer Res*. 2011; 30: 117. 10.1186/1756-9966-30-117

25. Nakata K, Nagai E, Miyasaka Y, et al.. The risk of lymph node metastasis in mucosal gastric carcinoma: especially for a mixture of differentiated and undifferentiated adenocarcinoma. *Hepatogastroenterology*. 2012; 59: 1855-8. 10.5754/hge10130

26. Huh CW, Jung DH, Kim JH, et al.. Signet ring cell mixed histology may show more aggressive behavior than other histologies in early gastric cancer. *J Surg Oncol*. 2017; 107: 124-9. 10.1002/jso.23261

27. Takizawa K, Ono H, Kakushima N, et al.. Risk of lymph node metastases from intramucosal gastric cancer in relation to histological types: how to manage the mixed histological type for endoscopic submucosal dissection. *Gastric Cancer*. 2013; 16: 531-6. 10.1007/s10120-012-0220-z

28. Sekiguchi M, Oda I, Taniguchi H, et al.. Risk stratification and predictive risk-scoring model for lymph node metastasis in early gastric cancer. *J Gastroenterol*. 2016; 51: 961-70. 10.1007/s00535-016-1180-6

29. Mikami K, Hirano Y, Futami K, et al.. Expansion of lymph node metastasis in mixed-type submucosal invasive gastric cancer. *Asian J Surg*. 2018; 41: 462-466. 10.1016/j.asjsur.2017.04.004

30. Komatsu S, Ichikawa D, Miyamae M, et al.. Histological mixed-type as an independent

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prognostic factor in stage I gastric carcinoma. *World J Gastroenterol*. 2015; 21: 549-55. 10.3748/wjg.v21.i2.549

31. Sentani K, Imai T, Kobayashi G, et al.. Histological diversity and molecular characteristics in gastric cancer: relation of cancer stem cell-related molecules and receptor tyrosine kinase molecules to mixed histological type and more histological patterns. *Gastric Cancer*. 2021; 24: 368-381. 10.1007/s10120-020-01133-w

32. Zheng HC, Li XH, Hara T, et al.. Mixed-type gastric carcinomas exhibit more aggressive features and indicate the histogenesis of carcinomas. *Virchows Arch*. 2008; 452: 525-34. 10.1007/s00428-007-0572-7

33. Park SY, Kook MC, Kim YW, et al.. Mixed-type gastric cancer and its association with high-frequency CpG island hypermethylation. *Virchows Arch*. 2010; 456: 625-33. 10.1007/s00428-010-0916-6

Figure legends

Fig.1. Flow chart summarizing the steps of identification for relevant studies.

Fig.2. Forest plots showing OR with 95% CI comparing MU with PU EGC for the risk of LNM.

Fig.3. Forest plots showing ORs with 95% CIs comparing MU with PU EGC for the risk of LNM. Studies are limited to those with patients from Japan.

Fig.4. Forest plots showing ORs with 95% CIs comparing MU with PU EGC for the risk of submucosal invasion.

Fig.5. Forest plots showing ORs with 95% CIs comparing MU with PU EGC for the risk of submucosal invasion. Studies are limited to those with patients from China.

Fig.6. Forest plots showing ORs with 95% CIs comparing MU with PU EGC for the risk of lymphovascular invasion.

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Fig.7. Forest plots showing ORs with 95% CIs comparing MU with PU EGC for the risk of lymphovascular invasion. Studies are limited to those with patients from Japan.

Fig.8. Forest plots showing ORs with 95% CIs comparing MU with PU EGC for the risk of the depressed (IIc) endoscopic growth patterns.

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Table 1. Clinical characteristics of patients from each of the included studies.

Author year	Country	MU/PU	LNM+	Depth (SM)	LVI+	Gross Type (Depressed)	Ulcerative findings	Score (NOS)
Hanaoka N. *et al.* [21], 2009	Japan	63/80	23/12		36/31			7
Ito H. *et al.* [24] , 2011	Japan	36/122	11/13					8
Nakata K. *et al.* [25], 2012	Japan	20/98	2/4					5
Huh CW. *et al.* [26], 2013	Korea	24/464	2/73	17/271	0/52	19/300		6
Takizawa K.*et al.* [27], 2013	Japan	42/184	8/11		2/4	40/165	21/81	7
Miyamae M. *et al.* [22] , 2016	Japan	45/38	14/3					7
Sekiguchi M. *et al.* [28], 2016	Japan	469/1202	96/103	252/437	120/118			6
Zhong Q. *et al.* [13] , 2018	China	17/69		9/20				6
Chen JN. *et al.* [19], 2020	China	144/572	39/110	99/272		122/448	23/61	7
Horiuchi Y. *et al.* [18] , 2020	Japan	525/900	104/66					7

Abbreviations: MU, undifferentiated-type-predominant mixed-type; PU, pure undifferentiated-type; LNM, lymph node metastasis; SM, submucosal; LVI, lymphovascular invasion; NOS, Newcastle-Ottawa Scale.

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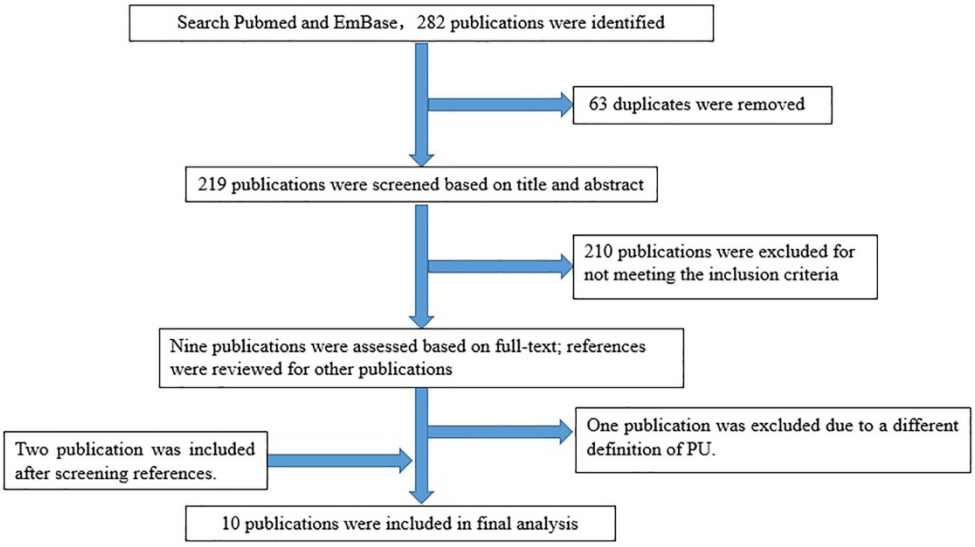


Fig.1.

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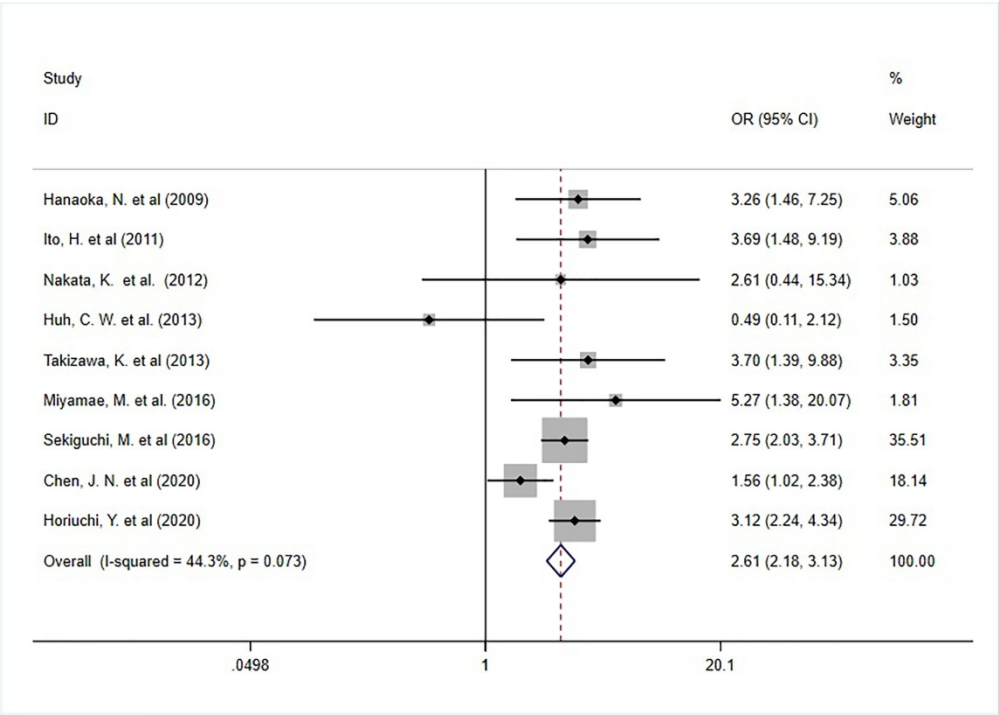


Fig.2.

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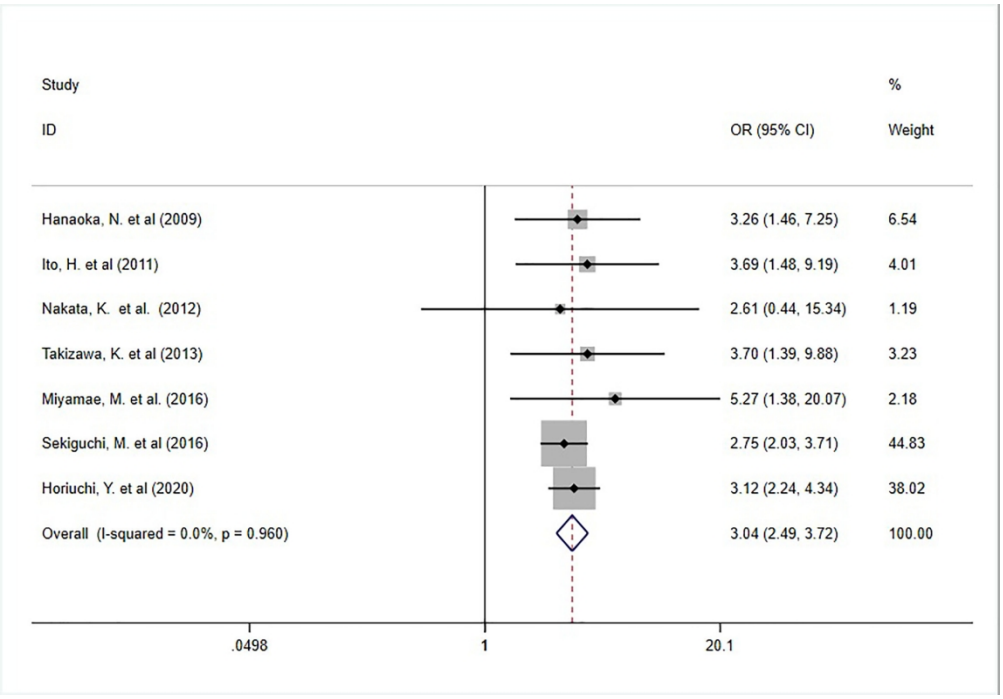


Fig.3.

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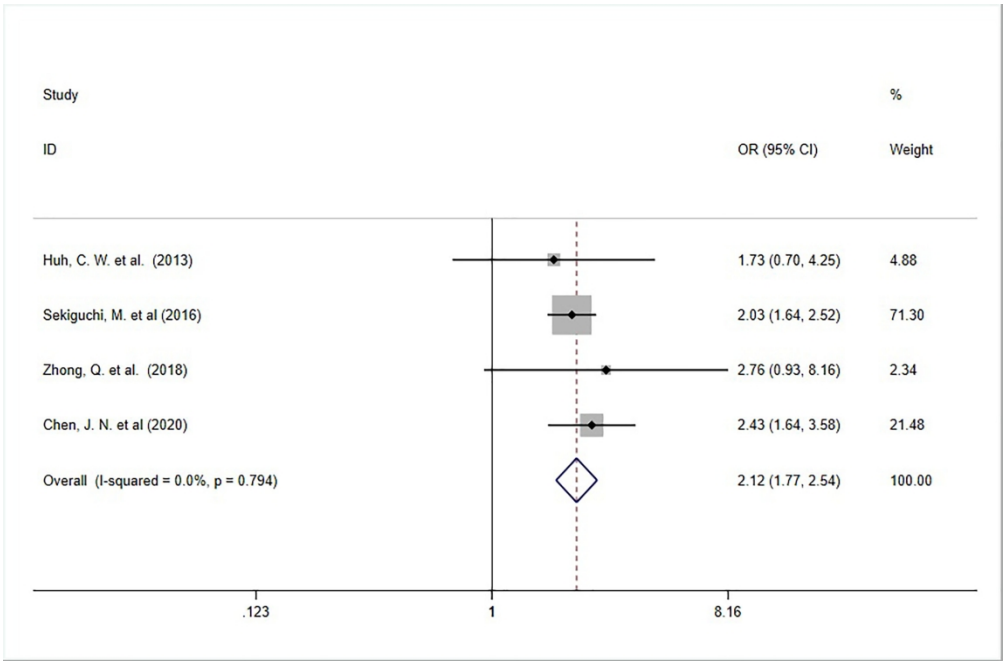


Fig.4.

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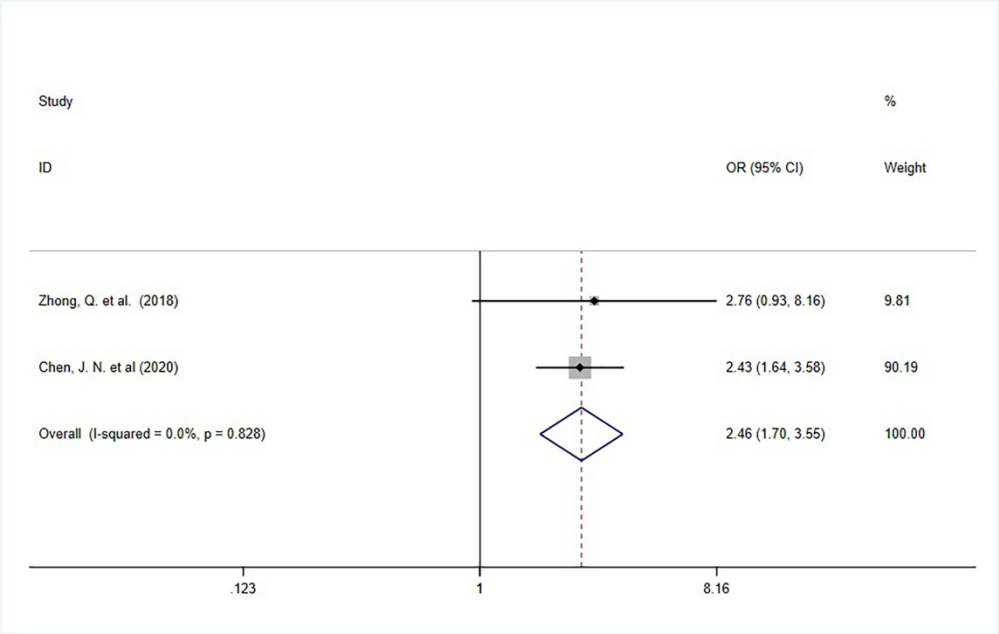


Fig.5.

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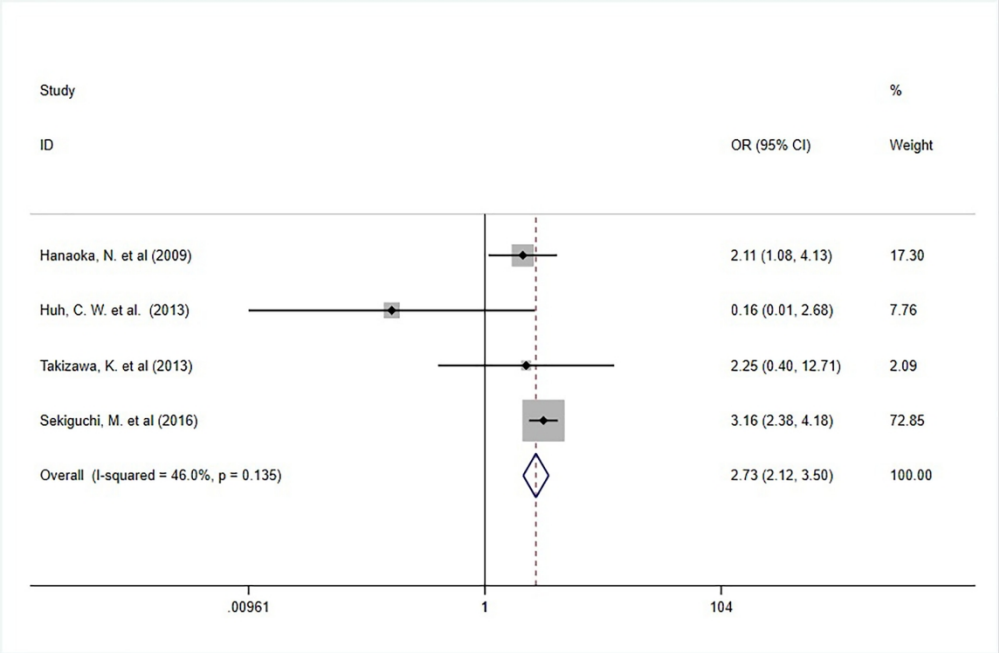


Fig.6.

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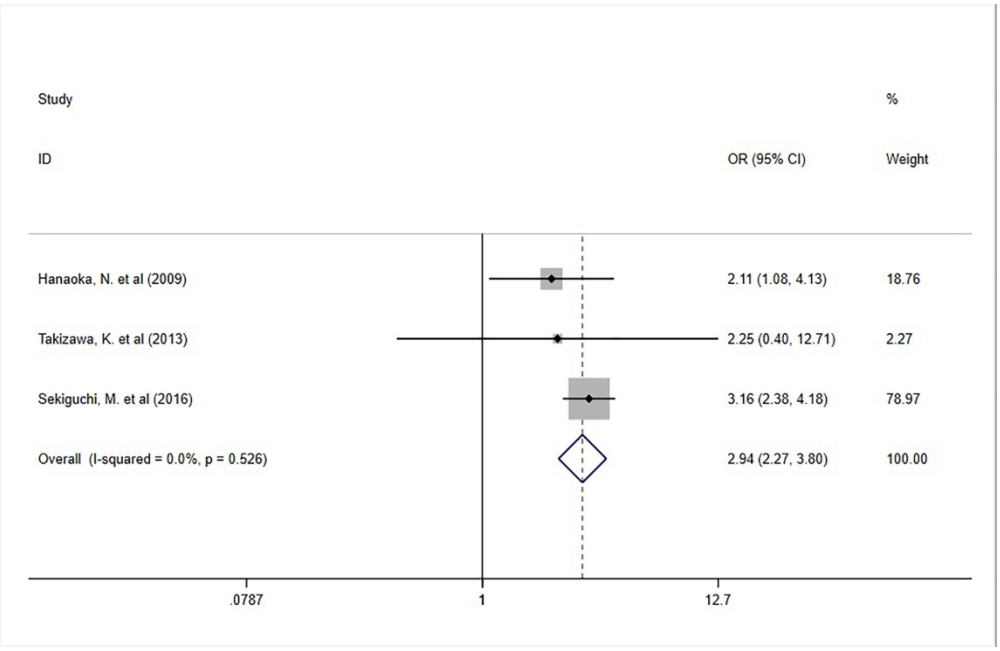


Fig.7.

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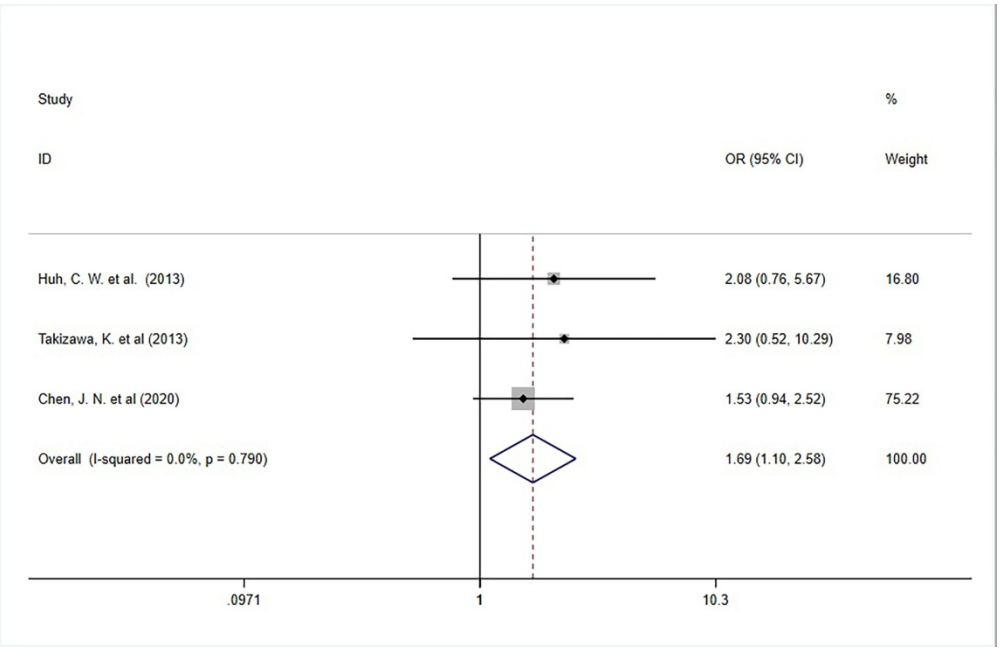


Fig.8.

251x161mm (300 x 300 DPI)

Search strategy in PubMed:
#1. (stomach neoplasma[MeSH Terms]) OR (early gastric cancer[Title/Abstract]) OR (EGC[Title/Abstract]) OR (early gastric carcinoma[Title/Abstract]) OR (mocosal[Title/Abstract]) OR (submocosal[Title/Abstract])
#2. (mixed type) OR (mixed histology) OR (undifferentiated) [Title/Abstract]
#3. #1 to #2
#4. #3 And English

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4-6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9-10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	15

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a systematic review and meta-analysis

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Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a systematic review and meta-analysis

Authors: Peng Yang[1,2#], Xiang-Dong Zheng[1,2#], Jun-Mei Wang[1,2], Wen-Bin Geng[1,2], Xiao-Yong Wang[1]*

1. Department of Gastroenterology, The Affiliated Changzhou No. 2 People's Hospital of Nanjing Medical University, 29 Xinglong Lane, Tianning District, Changzhou 213000, Jiangsu Province, China.

2. Graduate school, Dalian Medical University, 9 West Section of Lushun South Road, Lvshunkou District, Dalian 116000, Liaoning Province, China.

[#]The authors contributed equally to this work.

*Corresponding author: Xiao-Yong Wang, MD, From the Department of Gastroenterology, The Affiliated Changzhou No. 2 People's Hospital of Nanjing Medical University, 29 Xinglong Lane, Tianning District, Changzhou 213000, Jiangsu Province, China. wxy20009@126.com

ORICD ID: https://orcid.org/0000-0001-8401-4888

Authors' ORICD ID:

Xiang-Dong Zheng: https://orcid.org/0000-0003-1088-6872

Peng Yang: https://orcid.org/0000-0002-5501-0244

Jun-Mei Wang: https://orcid.org/0000-0002-2117-4144

Wen-Bin Geng: https://orcid.org/0000-0001-6770-7690

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Abstract

Background: Studies have shown that differentiated-predominant mixed-type early gastric cancer (EGC) is more aggressive than pure differentiated-type EGC. However, the biological behavior of undifferentiated-predominant mixed-type (MU) EGC and pure undifferentiated-type (PU) EGC are controversial. This study was conducted to compare the biological behavior of MU EGC and PU EGC.

Methods: A systematic review and meta-analysis of observational studies was conducted using literature published through PubMed and Embase from inception to November 9, 2021. Inclusion criteria were: (1) a direct or indirect comparison of MU and PU; (2) patients with EGC; (3) a specified outcome of lymph node metastasis (LNM), lymphovascular invasion, or submucosal invasion; and (4) the primary lesion was obtained. The literature search, data extraction, and quality assessment were performed by two independent reviewers. The meta-analysis was conducted with a random-effects model using the Mantel- Haenszel method.

Results: Twelve publications with 5644 patients were included. Patients with MU EGC had significantly higher risk of LNM (OR, 2.28; 95%CI, 1.72–3.03) and submucosal invasion (OR, 2.19; 95%CI, 1.90–2.52) compared to PU EGC patients. No difference was found between MU and PU EGC patients with respect to lymphovascular invasion risk (OR, 1.81; 95%CI, 0.84-3.87). After stratifying the data according to depth of tumor invasion, a significantly higher risk for LNM was associated with intramucosal MU EGC (OR, 2.56; 95%CI, 1.66–3.95) and submucosal MU EGC (OR, 2.63; 95%CI, 2.06–3.06). Submucosal

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MU EGC also had a significantly higher risk of lymphovascular invasion (OR, 2.40; 95%CI, 1.79-3.21) compared with submucosal PU EGC.

Discussion: Patients with MU EGC had an increased risk of submucosal invasion and LNM compared with PU EGC patients. MU patients with submucosal EGC also had an increased lymphovascular invasion risk compared to PU patients. Therefore, attention should be focused on the clinical management of patients with MU EGC.

Keywords: lymph node metastasis; lymphovascular invasion; submucosal invasion; undifferentiated-type; mixed-type; early gastric cancer

Strengths and limitations of this study:

- This is the first meta-analysis to compare the biological behavior between MU and PU EGC.
- All included studies were carried out in Asia.
- There might be some selection bias that cannot be completely excluded as all studies were retrospective.
- The number of studies included in this meta-analysis was limited and the total number of patients included was relatively small.

Declarations

Funding statement:

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Background

Gastric cancer (GC) has become the fifth most common type of cancer and the third leading cause of death due to cancer in the world, and there were more than one million new cases of GC diagnosed in 2018[1]. The stage of GC at the time of diagnosis dictates patient prognosis; the five-year overall survival rate of early gastric cancer (EGC) is 92.0-97.2%[2], whereas for advanced GC, the five-year overall survival rate is only 23-36%[3]. Therefore, early detection, diagnosis and treatment of GC are essential to optimize patient survival. EGC is defined as cancer tissue limited to the mucosa and submucosa, regardless of lymph node metastasis (LNM)[4]. Endoscopic resection including endoscopic mucosal resection and endoscopic submucosal dissection, as well as surgery are treatment options for EGC, whereas a gastrectomy with adequate lymphadenectomy and chemotherapy is the only therapeutic option for advanced GC[5]. Histological types can be divided into the following four groups according to the proportions of differentiated and undifferentiated components: pure differentiated-type (PD, differentiated components only), pure undifferentiated-type (PU, undifferentiated components only), differentiated-predominant mixed-type (MD, major differentiated and minor undifferentiated components), and undifferentiated-predominant mixed-type (MU, major undifferentiated and minor differentiated components). Previous studies have shown that MD EGC has a significantly higher risk of LNM and a higher rate of non-curative resection when compared to PD EGC[6-8]. However, the biological behavior of MU EGC and PU EGC is controversial. Certain reports have claimed that MU EGC has a significantly higher risk of LNM than PU EGC[9]. However, Chen *et al.* reported that MU EGC did not show an increase in the incidence of LNM compared to PU EGC[10].

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Additionally, according to the treatment guidelines of the Japanese Gastric Cancer Association[11], MU and PU EGC are collectively treated as undifferentiated-type EGC. Differences in the biological behavior between PU and MU have not been given much consideration, and it remains unknown whether they should be treated using the same ESD indication criteria. It is necessary to combine data from the studies available to maximize the power of this research in a relatively rare circumstance to address this issue. The above-mentioned considerations prompted the generation of this meta-analysis to reveal the biological behaviors in MU and PU EGC and determine if there are important differences that can improve the clinical approach to treatment.

Methods

This meta-analysis was completed according to the principles of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-P)[12].

Literature search

Studies from inception to November 9, 2021 were searched using the PubMed and Embase databases. Due to language limitations, only studies published in English were selected. Two independent evaluators (Peng Yang and Xiang-dong Zheng) used common key words associated with “early gastric cancer” and “mixed type” to search the databases. Details of the search strategy are shown in the supplementary material. The titles and abstracts of each article were reviewed to exclude irrelevant publications, and full texts were read to determine whether the remaining studies meet the inclusion criteria. References of the included articles

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were carefully checked to ensure no literature was unintentionally omitted. Disagreements between the two evaluators were resolved by a third evaluator (Xiao-yong Wang). The detailed retrieval strategy is presented in **Fig. 1**.

Eligibility criteria

Sufficient information was required from the studies to construct a 2 x 2 contingency table for further analysis, and available information was either extracted directly or calculated from the data contained by the studies. This meta-analysis included studies with the following criteria: (1) direct and indirect comparison of the relationship between MU and PU; (2) patients with EGC; (3) a specified outcome of LNM, lymphovascular invasion, submucosal invasion and/or ulcer findings; and (4) the primary lesion was obtained by open surgery or laparoscopic surgery. For articles with overlapping data of the same source population, only the largest report was included.

Data extraction

Pertinent information was extracted from each study independently by two reviewers (Peng Yang and Xiang-dong Zheng) who followed the same criteria. Results were compared, and disparities were settled by a third evaluator (Xiao-yong Wang). The essential study information included: name of the first author(s); date of publication; region from which the study population was derived. The necessary clinicopathologic factors included: LNM (yes or no); depth of tumor invasion (mucosal or submucosal invasion); lymphovascular invasion (yes or no); and ulcer findings (yes or no).

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Quality assessment

The quality of the included studies was assessed by two initial evaluators (Peng Yang and Xiang-dong Zheng) using the Newcastle-Ottawa scale (NOS). There major aspects including selection, comparability, and exposure/outcome were used to evaluate each paper. The final score ranged from 0 to 9, where 1 point was given for meeting one required item, and at most 2 points were given for comparability. Articles with a score ≥ 6 were considered high quality. Discrepancies were resolved by a third evaluator (Xiao-yong Wang).

Statistical analysis

Stata software, version 15.1 (StataCorp LLC 4905 Lakeway Drive, College Station, TX 77845 USA) was used to perform all statistical analyses.

The pooled rates were calculated for LNM, submucosal invasion, lymphovascular invasion, macroscopic aspect of the lesions, and unclear findings between undifferentiated-predominant mixed-type and pure undifferentiated-type EGC groups, which were expressed as odds ratios (ORs) with 95% confidence intervals (95% CIs). In addition to the overall comparisons, mucosal/submucosal invasion-based and region-based subgroup analyses were also implemented.

A random-effects model using the Mantel- Haenszel method was performed for the meta-analysis [Stata code: metan tdeath tnodeath cdeath cnodeath, or random label (namevar=id, yearvar=year) counts]. Publication bias was tested using both Begg's and Egger's tests [Stata code: gen logor=log(_ES); gen selogor=_selogES; metabias6 logor

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selogor, graph(begg)]. In addition, a sensitivity analysis was performed to assess the stability and reliability of the results [Stata code: metaninf tdeath tnodeath cdeath cnodeath, label (namevar=id, yearvar=year) random or].

Patient and public involvement

There was no direct patient or public involvement in the design or conduct of this review.

Results

Literature identification

As shown in **Fig. 1**, a total of 305 publications were retrieved through the initial literature search. Among these search results, 69 duplicates were removed, which left 236 for screening based on title and abstract. Two hundred and twenty-five were excluded for not meeting the inclusion criteria as stated in the methods section. Eleven publications were assessed based on the full text, and two additional publications were included after reading the references of these nine publications in detail[13,14]. One publication was excluded due to a different definition of PU (only undifferentiated components *vs.* more than 90% undifferentiated components)[15]. After the thorough screening, 12 publications[9,10,13,14,16-23] with 5644 patients were included for this meta-analysis, and the basic summarizing information from these articles is shown in **Table 1**.

Meta-analysis

The risk of LNM and subgroup analysis

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As shown in **Fig. 2,** 11 studies were used to compare PU and MU EGC regarding LNM risk. MU EGC was found to have an increased risk of LNM (OR, 2.28; 95% CI, 1.72–3.03) compared to PU EGC. When stratified according to region, a significantly increased risk for LNM associated with MU EGC was found in studies with patients from Japan (OR, 3.03; 95% CI, 2.48-3.71) and China (OR, 1.52; 95% CI, 1.14-2.02).

A subgroup analysis was performed to evaluate the LNM risk according to the depth of tumor invasion. This subgroup analysis revealed that mucosal MU EGC had a significantly higher LNM risk compared with mucosal PU EGC (OR, 2.56; 95% CI, 1.66–3.95), and submucosal MU EGC also had a significantly higher LNM risk compared with submucosal PU EGC (OR, 2.63; 95% CI, 2.06–3.06; **Fig. 3**).

The risk of submucosal invasion and subgroup analysis

Five studies were used to compare PU and MU EGC regarding submucosal invasion risk. MU EGC was found to be associated with an increased risk of submucosal invasion (OR, 2.19; 95% CI, 1.90–2.52; **Fig. 4**). When stratified according to region, a significantly higher submucosal invasion risk associated with MU EGC was found in both Japanese (OR, 2.16; 95% CI, 1.85-2.52) and Chinese (OR, 2.46; 95% CI, 1.71-3.55) population-based studies (**Fig.4**).

The risk of lymphovascular invasion and subgroup analysis

As indicated in **Fig. 5**, five studies were used to compare PU and MU EGC with respect to lymphovascular invasion risk, and no significant difference was found (OR, 1.81; 95% CI,

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0.84-3.87). When stratified by region, a significantly higher lymphovascular invasion risk associated with MU EGC was found in the Japanese-based studies (OR, 2.33; 95% CI, 1.27-4.28).

As shown in **Fig. 6,** when stratified by the depth of tumor invasion, the subgroup analysis revealed that submucosal MU EGC had a significantly higher lymphovascular invasion risk compared with submucosal PU EGC (OR, 2.40; 95% CI, 1.79-3.21), whereas intramucosal PU and intramucosal MU EGC were comparable (OR, 0.94; 95% CI, 0.18-4.81).

The risk of ulcer finding

Data regarding ulcer findings were also extracted to compare MU and PU EGC from three studies, but no statistically significant difference was found (OR, 1.30; 95% CI, 0.99-1.71; **Fig. 7**).

Sensitivity analysis and publication bias

To assess publication bias of the included studies, Begg's and Egger's tests were performed. There was no evidence of publication bias in our meta-analysis for risk of LNM (Begg's test: P = 0.755; Egger's test: P = 0.708; **Fig. 8**), or submucosal invasion (Begg's test: P = 0.806; Egger's test: P = 0.813). Therefore, it can be suggested that the publication bias for the included studies was relatively small. Furthermore, the results of our meta-analysis may represent real differences in the biological behavior between MU and PU EGC.

A sensitivity analysis was also performed by removing one study at one time. The results concerning LNM and submucosal invasion risk were not reversed with the sequential

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removal of each study. Therefore, it can be concluded that MU EGC is associated with a higher risk of LNM and submucosal invasion as compared to PU EGC.

Discussion

This meta-analysis confirmed that MU and PU EGC display different biological behaviors. Patients with MU EGC had a significantly higher risk of LNM and submucosal invasion compared with PU EGC patients. Regarding lymphovascular invasion, no statistically significant difference was found between PU and MU EGC patients. However, MU EGC was found to have an increased risk when compared to PU EGC only in studies involving patients with submucosal invasion. Consequently, more attention should be given to patients with MU EGC for clinical management of the disease. Careful assessment is essential for patients with MU EGC before they undergo a treatment procedure.

The biological mechanism whereby MU is more aggressive than PU EGC is inconclusive, but previous studies can provide insight and propose areas of research to be investigated. From the viewpoint of molecular pathology, the genomic instability subtype of GC has been shown to be associated with different histological types: the chromosomal instability subtype of GC was significantly associated with increased histological numbers and mixed-type GC, whereas the genomic stability subtype of GC showed a significant relationship with pure-type GC[24]. From the viewpoint of epigenetics, it has been reported that the elevated methylation levels of CpG islands, which inhibit the expression of tumor suppressor genes, may promote the development of mixed-type GC over diffuse-type or intestinal-type GC[25]. Taken together, genomic instability, the expression level of related genes, and elevated methylation levels of

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CpG islands might contribute to the more aggressive behavior of MU compared with pure differentiated-type EGC. Although the exact mechanisms remain unknown, additional articles exploring the mechanism of this aggressive biological behavior of MU EGC would provide valuable insight in the future.

Previous studies have reported the biological behaviors and clinicopathologic features that differ between PU and MU EGC, but controversies remain. This meta-analysis brings together all the available information to draw a more substantial conclusion. However, the present study has several limitations. Firstly, all included studies were carried out in Asia. Consequently, extending these results beyond Asia may be inaccurate and require further study with data from different races and regions. Secondly, all studies were retrospective. Therefore, there might be some selection bias that cannot be completely excluded. Thirdly, the number of studies included in this meta-analysis was limited, and the total number of patients included was relatively small. As a result, the statistical effectiveness of the meta-analysis is somewhat low. Nevertheless, despite these limitations, this is the first meta-analysis to compare the biological behavior between MU and PU EGC. Additional large-scale, multicenter studies are expected to validate these results.

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Conclusions

In conclusion, this study identified that patients with MU EGC had an increased risk of LNM and submucosal invasion compared with PU EGC patients. With respect to lymphovascular invasion, no difference was found between PU and MU EGC in the mucosal subgroup. However, MU EGC was found to have an increased risk when compared to PU EGC only in

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studies involving patients with submucosal invasion. Hence, we suggest that the lymph node dissection in EGC patients with PU may be appropriately reduced and the lymphadenectomy in EGC patients with MU may be appropriately raised. In future clinical practice, extra effort should be dedicated to further substantiate the outcomes of the present study. These results will help doctors to select the appropriate clinical management and therapy options for patients with MU EGC to improve their outcomes.

Contributors: PY and XZ contributed equally to this work. XW: conception and design of the study, critical revision, acquisition of data, analysis and interpretation of data, drafting the article, final approval; PY and XZ: acquisition of data, analysis and interpretation of data, drafting the article, final approval; JW: interpretation of data, revising the article, final approval; WG: interpretation of data, revising the article, final approval.

Competing interests: The authors declare no competing interests.

Data availability statement: All data relevant to the study are included in the article.

Ethics approval and consent to participate: This study does not require ethics approval as it is a review based on published studies.

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References

1. Bray F, Ferlay J, Soerjomataram I, et al. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2018; 68: 394-424. 10.3322/caac.21492

2. Hahn KY, Park CH, Lee YK, et al. Comparative study between endoscopic submucosal dissection and surgery in patients with early gastric cancer. *Surg Endosc*. 2018; 32: 73-86. 10.1007/s00464-017-5640-8

3. Cunningham D, Allum WH, Stenning SP, et al. Perioperative chemotherapy versus surgery alone for resectable gastroesophageal cancer. *N Engl J Med*. 2006; 355: 11-20. 10.1056/NEJMoa055531

4. Japanese Gastric Cancer Association. Japanese classification of gastric carcinoma: 3rd English edition. *Gastric Cancer*. 2011; 14: 101-12. 10.1007/s10120-011-0041-5

5. Smyth EC, Nilsson M, Grabsch HI, et al. Gastric cancer. *Lancet*. 2020; 396: 635-648. 10.1016/S0140-6736(20)31288-5

6. Min BH, Kim KM, Park CK, et al. Outcomes of endoscopic submucosal dissection for differentiated-type early gastric cancer with histological heterogeneity. *Gastric Cancer*. 2015 ; 18: 618-26. 10.1007/s10120-014-0378-7

7. Shim CN, Chung H, Park JC, et al. Early gastric cancer with mixed histology predominantly of differentiated type is a distinct subtype with different therapeutic outcomes of endoscopic resection. *Surg Endosc*. 2015; 29: 1787-94. 10.1007/s00464-014-3861-7

8. Horiuchi Y, Fujisaki J, Yamamoto N, et al. Undifferentiated-type component mixed with differentiated-type early gastric cancer is a significant risk factor for endoscopic non-curative

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resection. *Dig Endosc*. 2018; 30: 624-632. 10.1111/den.13059

9. Horiuchi Y, Ida S, Yamamoto N, et al. Feasibility of further expansion of the indications for endoscopic submucosal dissection in undifferentiated-type early gastric cancer. *Gastric Cancer*. 2020; 23: 285-292. 10.1007/s10120-019-01003-0

10. Chen JN, Wang QW, Zhang QW, et al. Poorly differentiated is more significant than signet ring cell component for lymph node metastasis in mixed-type early gastric cancer: a retrospective study from a large-volume hospital. *Surg Endosc*. 2020. 10.1007/s00464-020-07532-5

11. Japanese Gastric Cancer Association. Japanese gastric cancer treatment guidelines 2018 (5th edition). *Gastric Cancer*. 2021; 24: 1-21. 10.1007/s10120-020-01042-y

12. Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ*. 2015; 350: g7647. 10.1136/bmj.g7647

13. Hanaoka N, Tanabe S, Mikami T, et al. Mixed-histologic-type submucosal invasive gastric cancer as a risk factor for lymph node metastasis: feasibility of endoscopic submucosal dissection. *Endoscopy*. 2009; 41: 427-32. 10.1055/s-0029-1214495

14. Miyamae M, Komatsu S, Ichikawa D, et al. Histological mixed-type as an independent risk factor for nodal metastasis in submucosal gastric cancer. *Tumour Biol*. 2016; 37: 709-14. 10.1007/s13277-015-3864-6

15. Seo HS, Lee GE, Kang MG, et al. Mixed Histology Is a Risk Factor for Lymph Node Metastasis in Early Gastric Cancer. *J Surg Res*. 2019; 236: 271-277. 10.1016/j.jss.2018.11.055

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16. Zhong Q, Sun Q, Xu GF, et al. Differential analysis of lymph node metastasis in histological mixed-type early gastric carcinoma in the mucosa and submucosa. *World J Gastroenterol*. 2018; 24: 87-95. 10.3748/wjg.v24.i1.87

17. Ito H, Inoue H, Ikeda H, et al. Clinicopathological characteristics and treatment strategies in early gastric cancer: a retrospective cohort study. *J Exp Clin Cancer Res*. 2011; 30: 117. 10.1186/1756-9966-30-117

18. Nakata K, Nagai E, Miyasaka Y, et al. The risk of lymph node metastasis in mucosal gastric carcinoma: especially for a mixture of differentiated and undifferentiated adenocarcinoma. *Hepatogastroenterology*. 2012; 59: 1855-8. 10.5754/hge10130

19. Huh CW, Jung DH, Kim JH, et al. Signet ring cell mixed histology may show more aggressive behavior than other histologies in early gastric cancer. *J Surg Oncol*. 2017; 107: 124-9. 10.1002/jso.23261

20. Takizawa K, Ono H, Kakushima N, et al. Risk of lymph node metastases from intramucosal gastric cancer in relation to histological types: how to manage the mixed histological type for endoscopic submucosal dissection. *Gastric Cancer*. 2013; 16: 531-6. 10.1007/s10120-012-0220-z

21. Sekiguchi M, Oda I, Taniguchi H, et al. Risk stratification and predictive risk-scoring model for lymph node metastasis in early gastric cancer. *J Gastroenterol*. 2016; 51: 961-70. 10.1007/s00535-016-1180-6

22. Liu P, Li L, Wang J, Song H, et al. Lymph node metastasis risk factors and applicability of endoscopic submucosal dissection in mixed-type early gastric cancer in Chinese patients. J Gastrointest Oncol. 2021 12:1444-1453. 10.21037/jgo-21-462.

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23. Jin X, Wu W, Zhao J, et al. Clinical Features and Risk Factors for Lymph Node Metastasis in Early Signet Ring Cell Gastric Cancer. Front Oncol. 2021; 11:630675. 10.3389/fonc.2021.630675.

24. Sentani K, Imai T, Kobayashi G, et al. Histological diversity and molecular characteristics in gastric cancer: relation of cancer stem cell-related molecules and receptor tyrosine kinase molecules to mixed histological type and more histological patterns. *Gastric Cancer*. 2021; 24: 368-381. 10.1007/s10120-020-01133-w

25. Park SY, Kook MC, Kim YW, et al. Mixed-type gastric cancer and its association with high-frequency CpG island hypermethylation. *Virchows Arch*. 2010; 456: 625-33. 10.1007/s00428-010-0916-6

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Figure legends

Fig. 1. Flow chart summarizing the steps of identification for relevant studies.

Fig. 2. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis. Subgroup analysis was based on region.

Fig. 3. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis. Subgroup analysis was based on mucosal/submucosal invasion.

Fig. 4. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of submucosal invasion. Subgroup analysis was based on region.

Fig. 5. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymphovascular invasion. Subgroup analysis was based on region.

Fig. 6. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymphovascular invasion. Subgroup analysis was based on mucosal/submucosal invasion.

Fig. 7. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of ulcer findings.

Fig. 8. Funnel plot comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis.

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Table 1. Clinical characteristics of patients from each of the included studies.

Author year	Country	Men/Women; Age (years)	MU/PU	LNM+	Depth (SM)	LVI+	Ulcerative findings	Score (NOS)
Hanaoka N. *et al.* [13], 2009	Japan	271/105; 229<60/ 147≥60	63/80	23/12	All SM	36/31		7
Ito H. *et al.* [17], 2011	Japan	204/123; 163<65/ 164≥65	36/122	11/13	M+SM			8
Nakata K. *et al.* [18], 2012	Japan	194/86; 126≤60/ 154≥61	20/98	2/4	M+SM			5
Huh CW. *et al.* [19], 2013	Korea	301/187 59.1±10.6 51.7±12.0 51.2±11.7	24/464	2/73	17/271 M+SM	0/52		6
Takizawa K.*et al.* [20], 2013	Japan	240/170 61 (29–87)	42/184	8/11	All M	2/4	21/81	7
Miyamae M. *et al.* [14], 2016	Japan	169/70 69 (35-91)	45/38	14/3	All SM			7
Sekiguchi M. *et al.* [21], 2016	Japan	2006/1125 62 (23–88)	469/1202	96/103	252/437 M+SM	120/118		6
			252/437 217/765	80/72 16/31	All SM All M	119/115 1/3		
Zhong Q. *et al.* [16], 2018	China	206/92; 59.5±12.1 (18-86)	17/69		9/20 M+SM			6
Chen JN. *et al.* [10], 2020	China	1069/527; 62 (17–88)	144/572	39/110	99/272 M+SM		23/61	7
Horiuchi Y. *et al.* [9], 2020	Japan	711/714; 60 (19–91)	525/900	104/66	332/386 M+SM			7
			193/514 332/386	15/13 99/53	All M All SM	0/9	143/364	
Liu P. *et al.* [22], 2021	China	571/241; 60.9±10.5	141/136	50/33	M+SM			8
Jin X. *et al.* [23],	China	347/231; 411<65/	129/124	29/24	M+SM			6

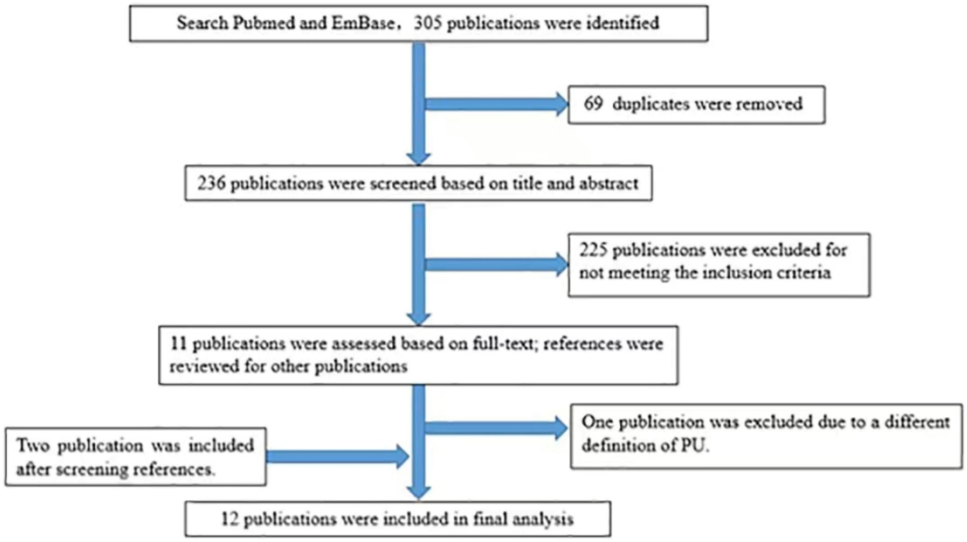
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2021	194≥65

Abbreviations: MU, undifferentiated-type-predominant mixed-type; PU, pure undifferentiated-type; LNM, lymph node metastasis; M, mucosal; SM, submucosal; LVI, lymphovascular invasion; NOS, Newcastle-Ottawa Scale.

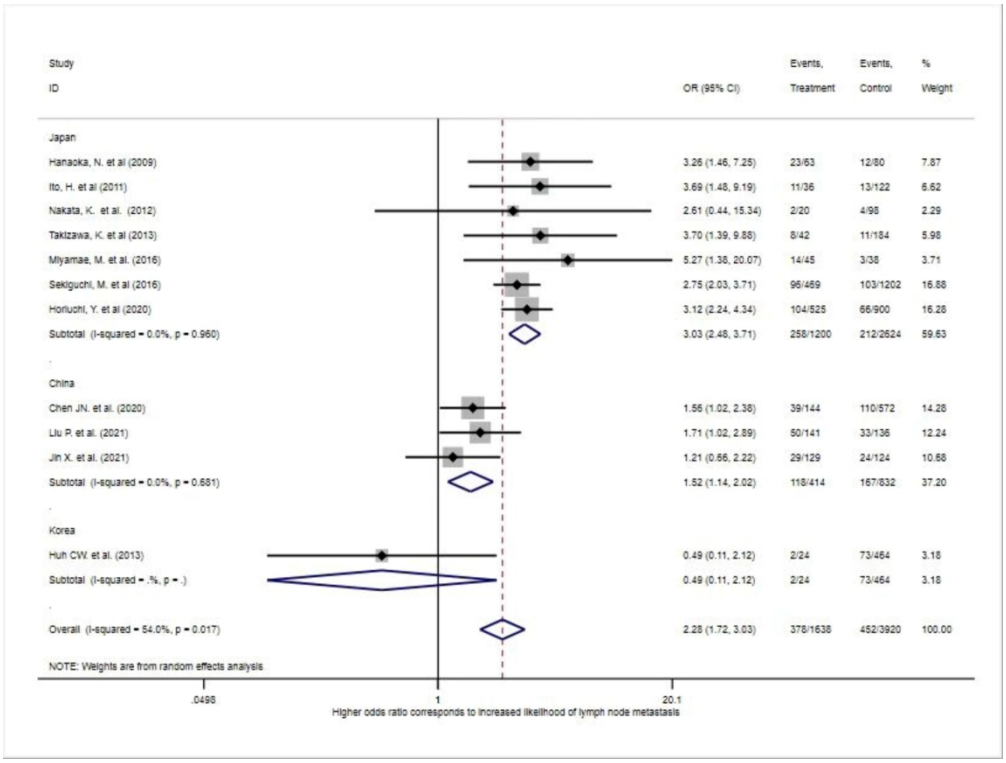
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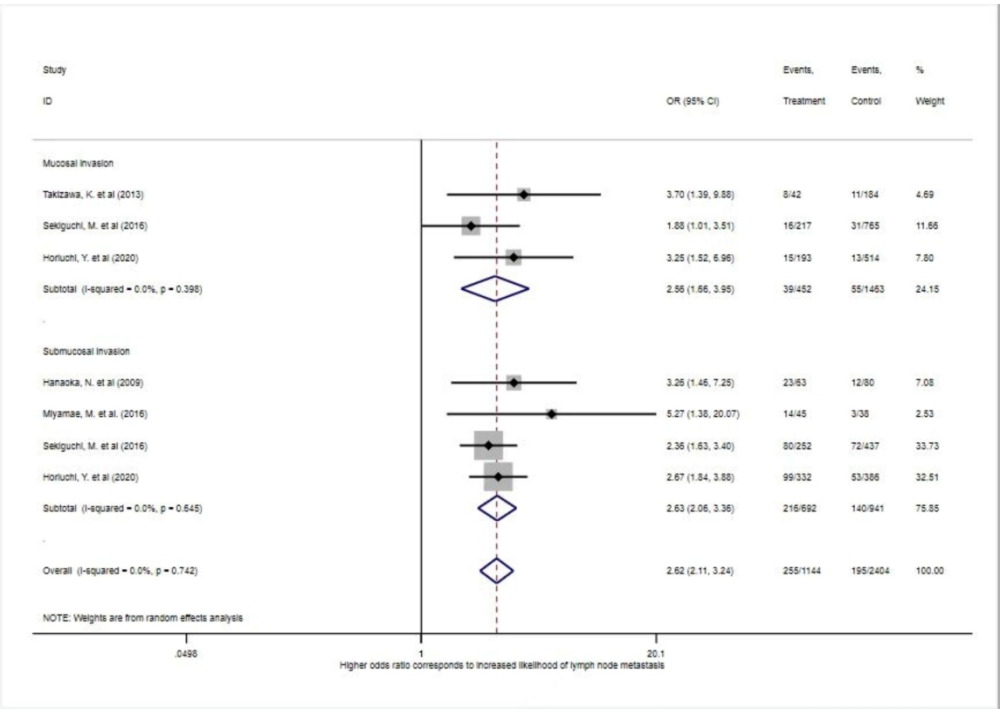
Flow chart summarizing the steps of identification for relevant studies.

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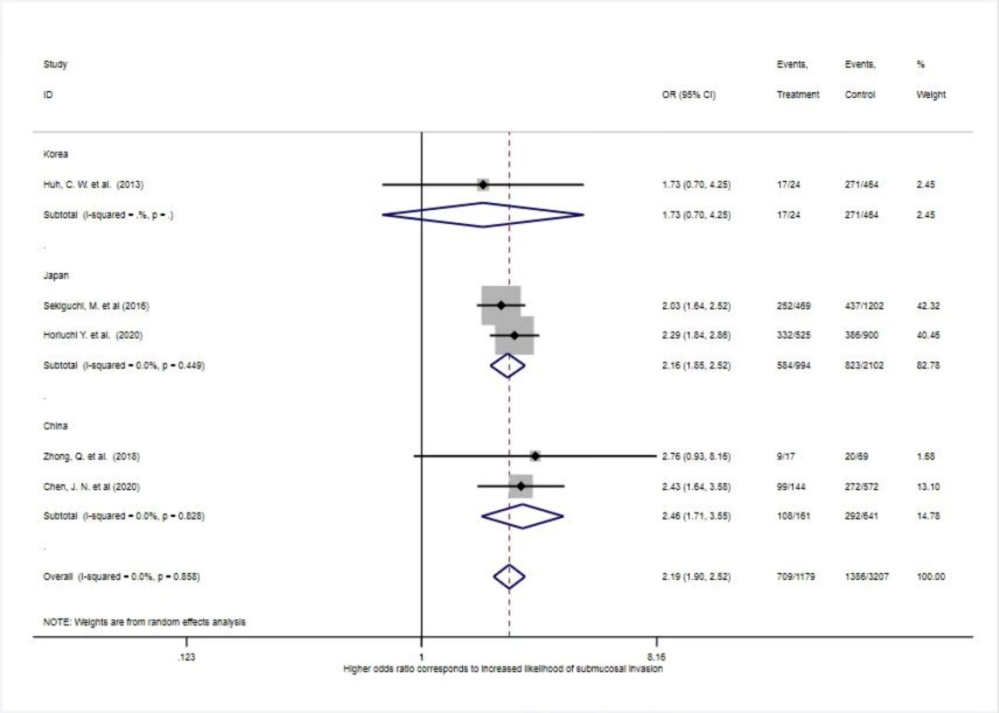
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis. Subgroup analysis was based on region.

285x215mm (300 x 300 DPI)



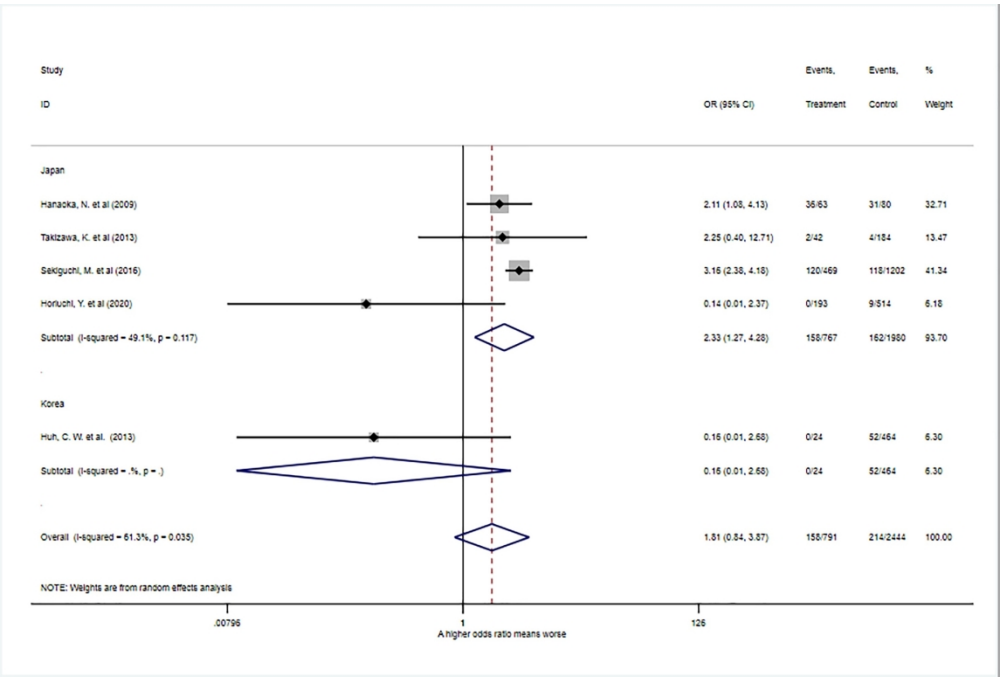
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis. Subgroup analysis was based on mucosal/submucosal invasion.

305x215mm (300 x 300 DPI)



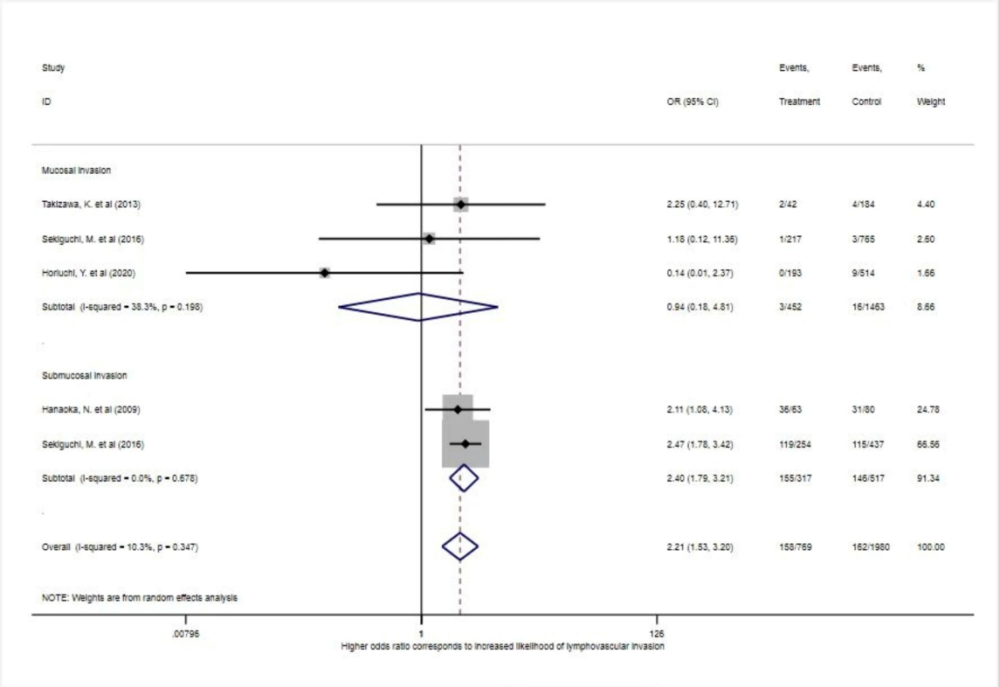
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of submucosal invasion. Subgroup analysis was based on region.

302x215mm (300 x 300 DPI)



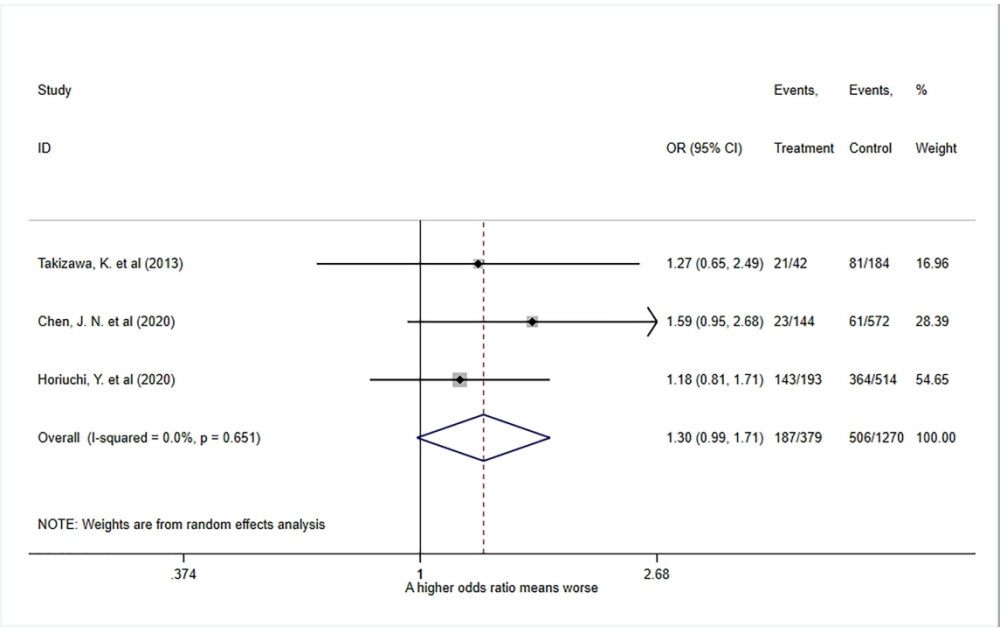
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk oflymphovascular invasion. Subgroup analysis was based on region.

240x161mm (300 x 300 DPI)



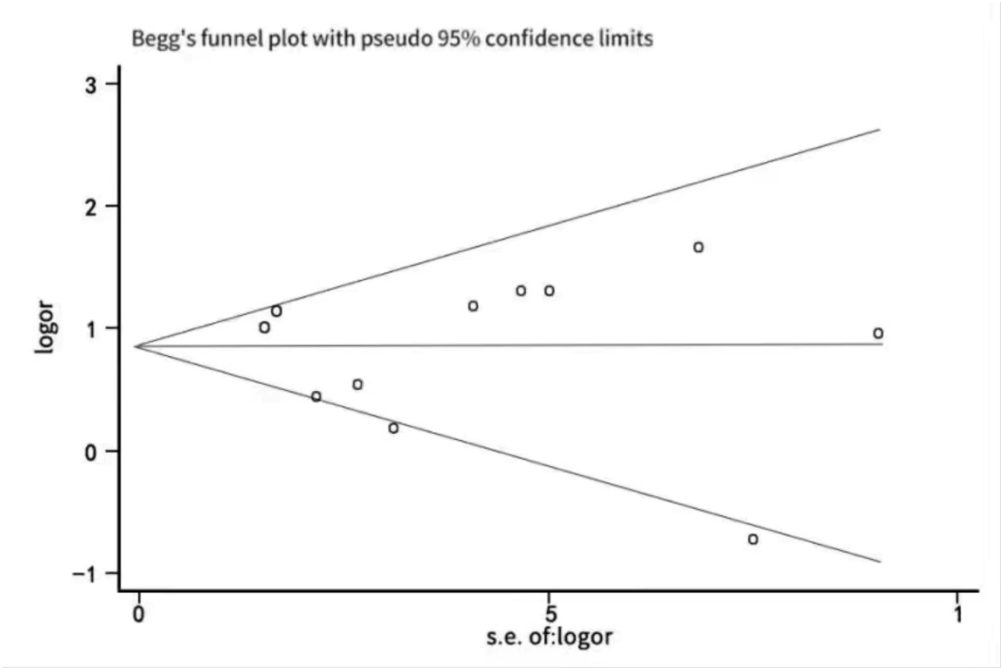
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymphovascular invasion. Subgroup analysis was based on mucosal/submucosal invasion.

312x215mm (300 x 300 DPI)



Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of ulcer findings.

259x161mm (300 x 300 DPI)



Funnel plot comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis.

282x188mm (300 x 300 DPI)

Search strategy in PubMed:

#1. (stomach neoplasma[MeSH Terms]) OR (early gastric cancer[Title/Abstract]) OR (EGC[Title/Abstract]) OR (early gastric carcinoma[Title/Abstract]) OR (mocosal[Title/Abstract]) OR (submocosal[Title/Abstract])

#2. (mixed type) OR (mixed histology) OR (undifferentiated) [Title/Abstract]

#3. #1 to #2

#4. #3 And English

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8-9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8-9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	13-14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	3-4

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Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a systematic review and meta-analysis

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Complete List of Authors:	Yang, Peng; The Affiliated Changzhou No. 2 People's Hospital of Nanjing Medical University, Department of Gastroenterology; Dalian Medical University Zheng, Xiang-Dong; The Affiliated Changzhou No. 2 People's Hospital of Nanjing Medical University, Department of Gastroenterology; Dalian Medical University Wang, Jun-Mei; The Affiliated Changzhou No. 2 People's Hospital of Nanjing Medical University, Department of Gastroenterology; Dalian Medical University Geng, Wen-Bin; The Affiliated Changzhou No. 2 People's Hospital of Nanjing Medical University, Department of Gastroenterology; Dalian Medical University wang, xiaoyong; The Affiliated Changzhou No 2 People's Hospital of Nanjing Medical University, gastroenterology
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Undifferentiated-predominant mixed-type early gastric cancer is more aggressive than pure undifferentiated-type: a systematic review and meta-analysis

Authors: Peng Yang[1,2#], Xiang-Dong Zheng[1,2#], Jun-Mei Wang[1,2], Wen-Bin Geng[1,2], Xiao-Yong Wang[1]*

1. Department of Gastroenterology, The Affiliated Changzhou No. 2 People's Hospital of Nanjing Medical University, 29 Xinglong Lane, Tianning District, Changzhou 213000, Jiangsu Province, China.

2. Graduate school, Dalian Medical University, 9 West Section of Lushun South Road, Lvshunkou District, Dalian 116000, Liaoning Province, China.

[#]The authors contributed equally to this work.

*Corresponding author: Xiao-Yong Wang, MD, From the Department of Gastroenterology, The Affiliated Changzhou No. 2 People's Hospital of Nanjing Medical University, 29 Xinglong Lane, Tianning District, Changzhou 213000, Jiangsu Province, China. wxy20009@126.com

ORICD ID: https://orcid.org/0000-0001-8401-4888

Authors' ORICD ID:

Xiang-Dong Zheng: https://orcid.org/0000-0003-1088-6872

Peng Yang: https://orcid.org/0000-0002-5501-0244

Jun-Mei Wang: https://orcid.org/0000-0002-2117-4144

Wen-Bin Geng: https://orcid.org/0000-0001-6770-7690

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Abstract

Background: Studies have shown that differentiated-predominant mixed-type early gastric cancer (EGC) is more aggressive than pure differentiated-type EGC. However, the biological behavior of undifferentiated-predominant mixed-type (MU) EGC and pure undifferentiated-type (PU) EGC are controversial. This study was conducted to compare the biological behavior of MU EGC and PU EGC.

Methods: A systematic review and meta-analysis of observational studies was conducted using literature published through PubMed and Embase from inception to November 9, 2021. Inclusion criteria were: (1) a direct or indirect comparison of MU and PU; (2) patients with EGC; (3) a specified outcome of lymph node metastasis (LNM), lymphovascular invasion, or submucosal invasion; and (4) the primary lesion was obtained. The literature search, data extraction, and quality assessment were performed by two independent reviewers. The meta-analysis was conducted with a random-effects model using the Mantel- Haenszel method.

Results: Twelve publications with 5644 patients were included. Patients with MU EGC had significantly higher risk of LNM (OR, 2.28; 95%CI, 1.72–3.03) and submucosal invasion (OR, 2.19; 95%CI, 1.90–2.52) compared to PU EGC patients. No difference was found between MU and PU EGC patients with respect to lymphovascular invasion risk (OR, 1.81; 95%CI, 0.84-3.87). After stratifying the data according to depth of tumor invasion, a significantly higher risk for LNM was associated with intramucosal MU EGC (OR, 2.56; 95%CI, 1.66–3.95) and submucosal MU EGC (OR, 2.63; 95%CI, 2.06–3.06). Submucosal

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MU EGC also had a significantly higher risk of lymphovascular invasion (OR, 2.40; 95%CI, 1.79-3.21) compared with submucosal PU EGC.

Discussion: Patients with MU EGC had an increased risk of submucosal invasion and LNM compared with PU EGC patients. MU patients with submucosal EGC also had an increased lymphovascular invasion risk compared to PU patients. Therefore, attention should be focused on the clinical management of patients with MU EGC.

Keywords: lymph node metastasis; lymphovascular invasion; submucosal invasion; undifferentiated-type; mixed-type; early gastric cancer

Strengths and limitations of this study:

- This is the first meta-analysis to compare the biological behavior between MU and PU EGC.
- All included studies were carried out in Asia.
- There might be some selection bias that cannot be completely excluded as all studies were retrospective.
- The number of studies included in this meta-analysis was limited and the total number of patients included was relatively small.

Declarations

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Background

Gastric cancer (GC) has become the fifth most common type of cancer and the third leading cause of death due to cancer in the world, and there were more than one million new cases of GC diagnosed in 2018[1]. The stage of GC at the time of diagnosis dictates patient prognosis; the five-year overall survival rate of early gastric cancer (EGC) is 92.0-97.2%[2], whereas for advanced GC, the five-year overall survival rate is only 23-36%[3]. Therefore, early detection, diagnosis and treatment of GC are essential to optimize patient survival. EGC is defined as cancer tissue limited to the mucosa and submucosa, regardless of lymph node metastasis (LNM)[4]. Endoscopic resection including endoscopic mucosal resection and endoscopic submucosal dissection, as well as surgery are treatment options for EGC, whereas a gastrectomy with adequate lymphadenectomy and chemotherapy is the only therapeutic option for advanced GC[5]. Histological types can be divided into the following four groups according to the proportions of differentiated and undifferentiated components: pure differentiated-type (PD, differentiated components only), pure undifferentiated-type (PU, undifferentiated components only), differentiated-predominant mixed-type (MD, major differentiated and minor undifferentiated components), and undifferentiated-predominant mixed-type (MU, major undifferentiated and minor differentiated components). Previous studies have shown that MD EGC has a significantly higher risk of LNM and a higher rate of non-curative resection when compared to PD EGC[6-8]. However, the biological behavior of MU EGC and PU EGC is controversial. Certain reports have claimed that MU EGC has a significantly higher risk of LNM than PU EGC[9]. However, Chen *et al.* reported that MU EGC did not show an increase in the incidence of LNM compared to PU EGC[10].

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Additionally, according to the treatment guidelines of the Japanese Gastric Cancer Association[11], MU and PU EGC are collectively treated as undifferentiated-type EGC. Differences in the biological behavior between PU and MU have not been given much consideration, and it remains unknown whether they should be treated using the same ESD indication criteria. It is necessary to combine data from the studies available to maximize the power of this research in a relatively rare circumstance to address this issue. The above-mentioned considerations prompted the generation of this meta-analysis to reveal the biological behaviors in MU and PU EGC and determine if there are important differences that can improve the clinical approach to treatment.

Methods

This meta-analysis was completed according to the principles of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-P)[12].

Literature search

Studies from inception to November 9, 2021 were searched using the PubMed and Embase databases. Due to language limitations, only studies published in English were selected. Two independent evaluators (Peng Yang and Xiang-dong Zheng) used common key words associated with “early gastric cancer” and “mixed type” to search the databases. Details of the search strategy are shown in the supplementary material. The titles and abstracts of each article were reviewed to exclude irrelevant publications, and full texts were read to determine whether the remaining studies meet the inclusion criteria. References of the included articles

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were carefully checked to ensure no literature was unintentionally omitted. Disagreements between the two evaluators were resolved by a third evaluator (Xiao-yong Wang). The detailed retrieval strategy is presented in **Fig. 1**.

Eligibility criteria

Sufficient information was required from the studies to construct a 2 x 2 contingency table for further analysis, and available information was either extracted directly or calculated from the data contained by the studies. This meta-analysis included studies with the following criteria: (1) direct and indirect comparison of the relationship between MU and PU; (2) patients with EGC; (3) a specified outcome of LNM, lymphovascular invasion, submucosal invasion and/or ulcer findings; and (4) the primary lesion was obtained by open surgery or laparoscopic surgery. For articles with overlapping data of the same source population, only the largest report was included.

Data extraction

Pertinent information was extracted from each study independently by two reviewers (Peng Yang and Xiang-dong Zheng) who followed the same criteria. Results were compared, and disparities were settled by a third evaluator (Xiao-yong Wang). The essential study information included: name of the first author(s); date of publication; region from which the study population was derived. The necessary clinicopathologic factors included: LNM (yes or no); depth of tumor invasion (mucosal or submucosal invasion); lymphovascular invasion (yes or no); and ulcer findings (yes or no).

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Quality assessment

The quality of the included studies was assessed by two initial evaluators (Peng Yang and Xiang-dong Zheng) using the Newcastle-Ottawa scale (NOS). There major aspects including selection, comparability, and exposure/outcome were used to evaluate each paper. The final score ranged from 0 to 9, where 1 point was given for meeting one required item, and at most 2 points were given for comparability. Articles with a score ≥ 6 were considered high quality. Discrepancies were resolved by a third evaluator (Xiao-yong Wang).

Statistical analysis

Stata software, version 15.1 (StataCorp LLC 4905 Lakeway Drive, College Station, TX 77845 USA) was used to perform all statistical analyses.

The pooled rates were calculated for LNM, submucosal invasion, lymphovascular invasion, macroscopic aspect of the lesions, and unclear findings between undifferentiated-predominant mixed-type and pure undifferentiated-type EGC groups, which were expressed as odds ratios (ORs) with 95% confidence intervals (95% CIs). In addition to the overall comparisons, mucosal/submucosal invasion-based and region-based (only articles from China and Japan which have enough data can be used for region-based subgroup analysis) subgroup analyses were also implemented.

A random-effects model using the Mantel- Haenszel method was performed for the meta-analysis [Stata code: metan tdeath tnodeath cdeath cnodeath, or random label (namevar=id, yearvar=year) counts]. Publication bias was tested using both Begg's and

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Egger's tests [Stata code: gen logor=log(_ES); gen selogor=_selogES; metabias6 logor selogor, graph(begg)]. In addition, a sensitivity analysis was performed to assess the stability and reliability of the results [Stata code: metaninf tdeath tnodeath cdeath cnodeath, label (namevar=id, yearvar=year) random or].

Patient and public involvement

There was no direct patient or public involvement in the design or conduct of this review.

Results

Literature identification

As shown in **Fig. 1**, a total of 305 publications were retrieved through the initial literature search. Among these search results, 69 duplicates were removed, which left 236 for screening based on title and abstract. Two hundred and twenty-five were excluded for not meeting the inclusion criteria as stated in the methods section. Eleven publications were assessed based on the full text, and two additional publications were included after reading the references of these nine publications in detail[13,14]. One publication was excluded due to a different definition of PU (only undifferentiated components *vs.* more than 90% undifferentiated components)[15]. After the thorough screening, 12 publications[9,10,13,14,16-23] with 5644 patients were included for this meta-analysis, and the basic summarizing information from these articles is shown in **Table 1**.

Meta-analysis

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The risk of LNM and subgroup analysis

As shown in **Fig. 2,** 11 studies were used to compare PU and MU EGC regarding LNM risk. MU EGC was found to have an increased risk of LNM (OR, 2.28; 95% CI, 1.72–3.03) compared to PU EGC. When stratified according to region, a significantly increased risk for LNM associated with MU EGC was found in studies with patients from Japan (OR, 3.03; 95% CI, 2.48-3.71) and China (OR, 1.52; 95% CI, 1.14-2.02).

A subgroup analysis was performed to evaluate the LNM risk according to the depth of tumor invasion. This subgroup analysis revealed that mucosal MU EGC had a significantly higher LNM risk compared with mucosal PU EGC (OR, 2.56; 95% CI, 1.66–3.95), and submucosal MU EGC also had a significantly higher LNM risk compared with submucosal PU EGC (OR, 2.63; 95% CI, 2.06–3.06; **Fig. 3**).

The risk of submucosal invasion and subgroup analysis

Five studies were used to compare PU and MU EGC regarding submucosal invasion risk. MU EGC was found to be associated with an increased risk of submucosal invasion (OR, 2.19; 95% CI, 1.90–2.52; **Fig. 4**). When stratified according to region, a significantly higher submucosal invasion risk associated with MU EGC was found in both Japanese (OR, 2.16; 95% CI, 1.85-2.52) and Chinese (OR, 2.46; 95% CI, 1.71-3.55) population-based studies (**Fig.4**).

The risk of lymphovascular invasion and subgroup analysis

As indicated in **Fig. 5**, five studies were used to compare PU and MU EGC with respect to

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lymphovascular invasion risk, and no significant difference was found (OR, 1.81; 95% CI, 0.84-3.87). When stratified by region, a significantly higher lymphovascular invasion risk associated with MU EGC was found in the Japanese-based studies (OR, 2.33; 95% CI, 1.27-4.28).

As shown in **Fig. 6,** when stratified by the depth of tumor invasion, the subgroup analysis revealed that submucosal MU EGC had a significantly higher lymphovascular invasion risk compared with submucosal PU EGC (OR, 2.40; 95% CI, 1.79-3.21), whereas intramucosal PU and intramucosal MU EGC were comparable (OR, 0.94; 95% CI, 0.18-4.81).

The risk of ulcer finding

Data regarding ulcer findings were also extracted to compare MU and PU EGC from three studies, but no statistically significant difference was found (OR, 1.30; 95% CI, 0.99-1.71; **Fig. 7**).

Sensitivity analysis and publication bias

To assess publication bias of the included studies, Begg's and Egger's tests were performed. There was no evidence of publication bias in our meta-analysis for risk of LNM (Begg's test: P = 0.755; Egger's test: P = 0.708; **Fig. 8**), or submucosal invasion (Begg's test: P = 0.806; Egger's test: P = 0.813). Therefore, it can be suggested that the publication bias for the included studies was relatively small. Furthermore, the results of our meta-analysis may represent real differences in the biological behavior between MU and PU EGC.

A sensitivity analysis was also performed by removing one study at one time. The results

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concerning LNM and submucosal invasion risk were not reversed with the sequential removal of each study. Therefore, it can be concluded that MU EGC is associated with a higher risk of LNM and submucosal invasion as compared to PU EGC.

Discussion

This meta-analysis confirmed that MU and PU EGC display different biological behaviors. Patients with MU EGC had a significantly higher risk of LNM and submucosal invasion compared with PU EGC patients. Regarding lymphovascular invasion, no statistically significant difference was found between PU and MU EGC patients. However, MU EGC was found to have an increased risk when compared to PU EGC only in studies involving patients with submucosal invasion. Consequently, more attention should be given to patients with MU EGC for clinical management of the disease. Careful assessment is essential for patients with MU EGC before they undergo a treatment procedure.

The biological mechanism whereby MU is more aggressive than PU EGC is inconclusive, but previous studies can provide insight and propose areas of research to be investigated. From the viewpoint of molecular pathology, the genomic instability subtype of GC has been shown to be associated with different histological types: the chromosomal instability subtype of GC was significantly associated with increased histological numbers and mixed-type GC, whereas the genomic stability subtype of GC showed a significant relationship with pure-type GC[24]. From the viewpoint of epigenetics, it has been reported that the elevated methylation levels of CpG islands, which inhibit the expression of tumor suppressor genes, may promote the development of mixed-type GC over diffuse-type or intestinal-type GC[25]. Taken together,

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genomic instability, the expression level of related genes, and elevated methylation levels of CpG islands might contribute to the more aggressive behavior of MU compared with pure differentiated-type EGC. Although the exact mechanisms remain unknown, additional articles exploring the mechanism of this aggressive biological behavior of MU EGC would provide valuable insight in the future.

Previous studies have reported the biological behaviors and clinicopathologic features that differ between PU and MU EGC, but controversies remain. This meta-analysis brings together all the available information to draw a more substantial conclusion. However, the present study has several limitations. Firstly, all included studies were carried out in Asia. Consequently, extending these results beyond Asia may be inaccurate and require further study with data from different races and regions. Secondly, all studies were retrospective. Therefore, there might be some selection bias that cannot be completely excluded. Thirdly, the number of studies included in this meta-analysis was limited, and the total number of patients included was relatively small. As a result, the statistical effectiveness of the meta-analysis is somewhat low. Nevertheless, despite these limitations, this is the first meta-analysis to compare the biological behavior between MU and PU EGC. Additional large-scale, multicenter studies are expected to validate these results.

Conclusions

In conclusion, this study identified that patients with MU EGC had an increased risk of LNM and submucosal invasion compared with PU EGC patients. With respect to lymphovascular invasion, no difference was found between PU and MU EGC in the mucosal subgroup.

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However, MU EGC was found to have an increased risk when compared to PU EGC only in studies involving patients with submucosal invasion. Hence, we suggest that the lymph node dissection in EGC patients with PU may be appropriately reduced and the lymphadenectomy in EGC patients with MU may be appropriately raised. In future clinical practice, extra effort should be dedicated to further substantiate the outcomes of the present study. These results will help doctors to select the appropriate clinical management and therapy options for patients with MU EGC to improve their outcomes.

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Competing interests: The authors declare no competing interests.

Data availability statement: All data relevant to the study are included in the article.

Ethics approval and consent to participate: This study does not require ethics approval as it is a review based on published studies.

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References

1. Bray F, Ferlay J, Soerjomataram I, et al. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2018; 68: 394-424. 10.3322/caac.21492

2. Hahn KY, Park CH, Lee YK, et al. Comparative study between endoscopic submucosal dissection and surgery in patients with early gastric cancer. *Surg Endosc*. 2018; 32: 73-86. 10.1007/s00464-017-5640-8

3. Cunningham D, Allum WH, Stenning SP, et al. Perioperative chemotherapy versus surgery alone for resectable gastroesophageal cancer. *N Engl J Med*. 2006; 355: 11-20. 10.1056/NEJMoa055531

4. Japanese Gastric Cancer Association. Japanese classification of gastric carcinoma: 3rd English edition. *Gastric Cancer*. 2011; 14: 101-12. 10.1007/s10120-011-0041-5

5. Smyth EC, Nilsson M, Grabsch HI, et al. Gastric cancer. *Lancet*. 2020; 396: 635-648. 10.1016/S0140-6736(20)31288-5

6. Min BH, Kim KM, Park CK, et al. Outcomes of endoscopic submucosal dissection for differentiated-type early gastric cancer with histological heterogeneity. *Gastric Cancer*. 2015 ; 18: 618-26. 10.1007/s10120-014-0378-7

7. Shim CN, Chung H, Park JC, et al. Early gastric cancer with mixed histology predominantly of differentiated type is a distinct subtype with different therapeutic outcomes of endoscopic resection. *Surg Endosc*. 2015; 29: 1787-94. 10.1007/s00464-014-3861-7

8. Horiuchi Y, Fujisaki J, Yamamoto N, et al. Undifferentiated-type component mixed with differentiated-type early gastric cancer is a significant risk factor for endoscopic non-curative

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resection. *Dig Endosc*. 2018; 30: 624-632. 10.1111/den.13059

9. Horiuchi Y, Ida S, Yamamoto N, et al. Feasibility of further expansion of the indications for endoscopic submucosal dissection in undifferentiated-type early gastric cancer. *Gastric Cancer*. 2020; 23: 285-292. 10.1007/s10120-019-01003-0

10. Chen JN, Wang QW, Zhang QW, et al. Poorly differentiated is more significant than signet ring cell component for lymph node metastasis in mixed-type early gastric cancer: a retrospective study from a large-volume hospital. *Surg Endosc*. 2020. 10.1007/s00464-020-07532-5

11. Japanese Gastric Cancer Association. Japanese gastric cancer treatment guidelines 2018 (5th edition). *Gastric Cancer*. 2021; 24: 1-21. 10.1007/s10120-020-01042-y

12. Shamseer L, Moher D, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ*. 2015; 350: g7647. 10.1136/bmj.g7647

13. Hanaoka N, Tanabe S, Mikami T, et al. Mixed-histologic-type submucosal invasive gastric cancer as a risk factor for lymph node metastasis: feasibility of endoscopic submucosal dissection. *Endoscopy*. 2009; 41: 427-32. 10.1055/s-0029-1214495

14. Miyamae M, Komatsu S, Ichikawa D, et al. Histological mixed-type as an independent risk factor for nodal metastasis in submucosal gastric cancer. *Tumour Biol*. 2016; 37: 709-14. 10.1007/s13277-015-3864-6

15. Seo HS, Lee GE, Kang MG, et al. Mixed Histology Is a Risk Factor for Lymph Node Metastasis in Early Gastric Cancer. *J Surg Res*. 2019; 236: 271-277. 10.1016/j.jss.2018.11.055

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16. Zhong Q, Sun Q, Xu GF, et al. Differential analysis of lymph node metastasis in histological mixed-type early gastric carcinoma in the mucosa and submucosa. *World J Gastroenterol*. 2018; 24: 87-95. 10.3748/wjg.v24.i1.87

17. Ito H, Inoue H, Ikeda H, et al. Clinicopathological characteristics and treatment strategies in early gastric cancer: a retrospective cohort study. *J Exp Clin Cancer Res*. 2011; 30: 117. 10.1186/1756-9966-30-117

18. Nakata K, Nagai E, Miyasaka Y, et al. The risk of lymph node metastasis in mucosal gastric carcinoma: especially for a mixture of differentiated and undifferentiated adenocarcinoma. *Hepatogastroenterology*. 2012; 59: 1855-8. 10.5754/hge10130

19. Huh CW, Jung DH, Kim JH, et al. Signet ring cell mixed histology may show more aggressive behavior than other histologies in early gastric cancer. *J Surg Oncol*. 2017; 107: 124-9. 10.1002/jso.23261

20. Takizawa K, Ono H, Kakushima N, et al. Risk of lymph node metastases from intramucosal gastric cancer in relation to histological types: how to manage the mixed histological type for endoscopic submucosal dissection. *Gastric Cancer*. 2013; 16: 531-6. 10.1007/s10120-012-0220-z

21. Sekiguchi M, Oda I, Taniguchi H, et al. Risk stratification and predictive risk-scoring model for lymph node metastasis in early gastric cancer. *J Gastroenterol*. 2016; 51: 961-70. 10.1007/s00535-016-1180-6

22. Liu P, Li L, Wang J, Song H, et al. Lymph node metastasis risk factors and applicability of endoscopic submucosal dissection in mixed-type early gastric cancer in Chinese patients. J Gastrointest Oncol. 2021 12:1444-1453. 10.21037/jgo-21-462.

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23. Jin X, Wu W, Zhao J, et al. Clinical Features and Risk Factors for Lymph Node Metastasis in Early Signet Ring Cell Gastric Cancer. Front Oncol. 2021; 11:630675. 10.3389/fonc.2021.630675.

24. Sentani K, Imai T, Kobayashi G, et al. Histological diversity and molecular characteristics in gastric cancer: relation of cancer stem cell-related molecules and receptor tyrosine kinase molecules to mixed histological type and more histological patterns. *Gastric Cancer*. 2021; 24: 368-381. 10.1007/s10120-020-01133-w

25. Park SY, Kook MC, Kim YW, et al. Mixed-type gastric cancer and its association with high-frequency CpG island hypermethylation. *Virchows Arch*. 2010; 456: 625-33. 10.1007/s00428-010-0916-6

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Figure legends

Fig. 1. Flow chart summarizing the steps of identification for relevant studies.

Fig. 2. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis. Subgroup analysis was based on region.

Fig. 3. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis. Subgroup analysis was based on mucosal/submucosal invasion.

Fig. 4. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of submucosal invasion. Subgroup analysis was based on region.

Fig. 5. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymphovascular invasion. Subgroup analysis was based on region.

Fig. 6. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymphovascular invasion. Subgroup analysis was based on mucosal/submucosal invasion.

Fig. 7. Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of ulcer findings.

Fig. 8. Funnel plot comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis.

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Table 1. Clinical characteristics of patients from each of the included studies.

Author year	Country	Men/Women; Age (years)	MU/PU	LNM+	Depth (SM)	LVI+	Ulcerative findings	Score (NOS)
Hanaoka N. *et al.* [13], 2009	Japan	271/105; 229<60/ 147≥60	63/80	23/12	All SM	36/31		7
Ito H. *et al.* [17], 2011	Japan	204/123; 163<65/ 164≥65	36/122	11/13	M+SM			8
Nakata K. *et al.* [18], 2012	Japan	194/86; 126≤60/ 154≥61	20/98	2/4	M+SM			5
Huh CW. *et al.* [19], 2013	Korea	301/187 59.1±10.6 51.7±12.0 51.2±11.7	24/464	2/73	17/271 M+SM	0/52		6
Takizawa K.*et al.* [20], 2013	Japan	240/170 61 (29–87)	42/184	8/11	All M	2/4	21/81	7
Miyamae M. *et al.* [14], 2016	Japan	169/70 69 (35-91)	45/38	14/3	All SM			7
Sekiguchi M. *et al.* [21], 2016	Japan	2006/1125 62 (23–88)	469/1202	96/103	252/437 M+SM	120/118		6
			252/437 217/765	80/72 16/31	All SM All M	119/115 1/3		
Zhong Q. *et al.* [16], 2018	China	206/92; 59.5±12.1 (18-86)	17/69		9/20 M+SM			6
Chen JN. *et al.* [10], 2020	China	1069/527; 62 (17–88)	144/572	39/110	99/272 M+SM		23/61	7
Horiuchi Y. *et al.* [9], 2020	Japan	711/714; 60 (19–91)	525/900	104/66	332/386 M+SM			7
			193/514 332/386	15/13 99/53	All M All SM	0/9	143/364	
Liu P. *et al.* [22], 2021	China	571/241; 60.9±10.5	141/136	50/33	M+SM			8
Jin X. *et al.* [23],	China	347/231; 411<65/	129/124	29/24	M+SM			6

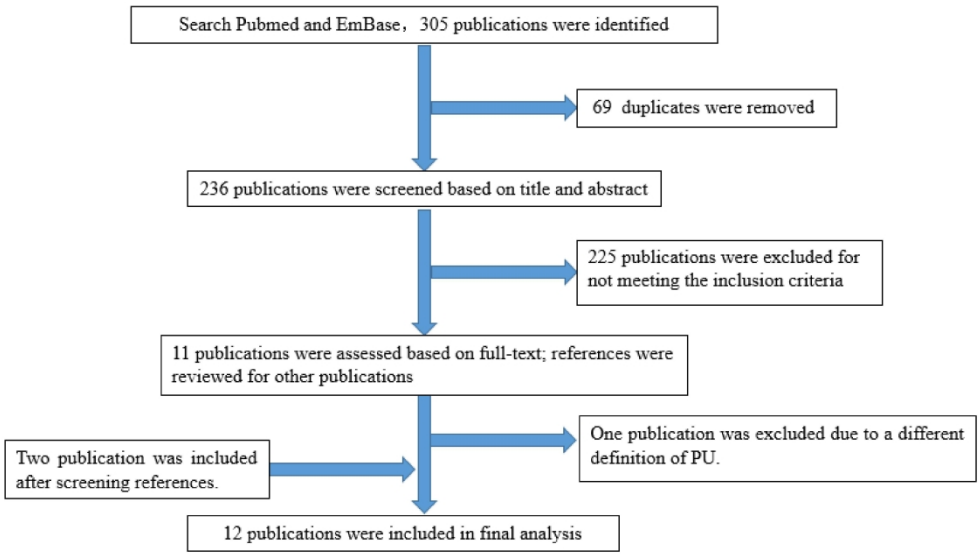
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2021	194≥65

Abbreviations: MU, undifferentiated-predominant mixed-type; PU, pure undifferentiated-type; LNM, lymph node metastasis; M, mucosal; SM, submucosal; LVI, lymphovascular invasion; NOS, Newcastle-Ottawa Scale.

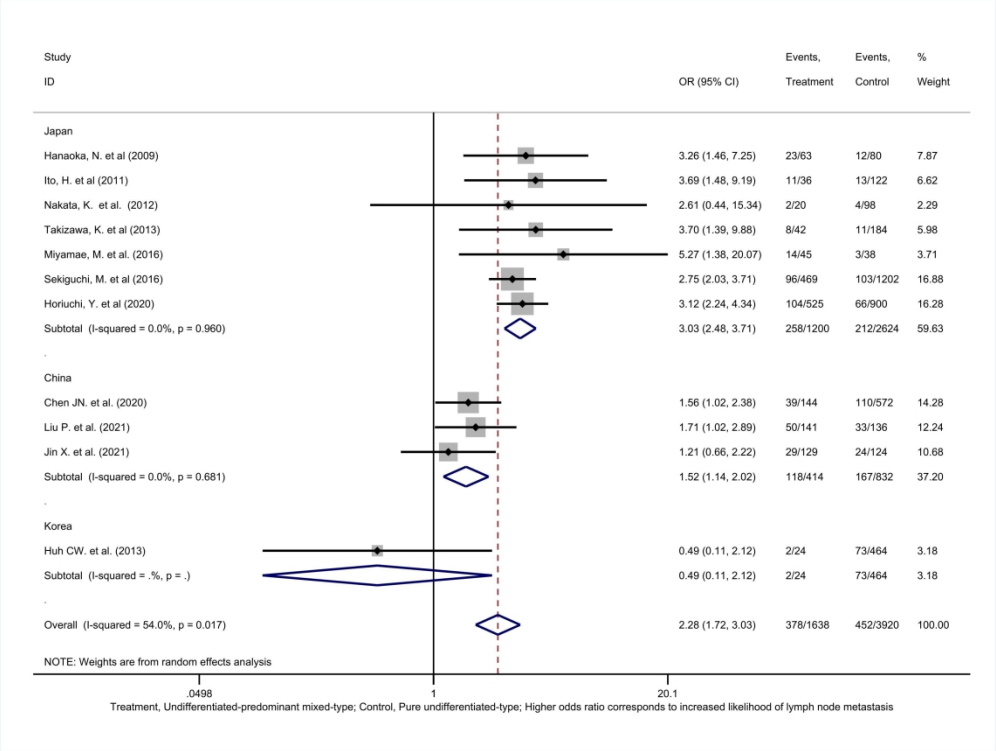
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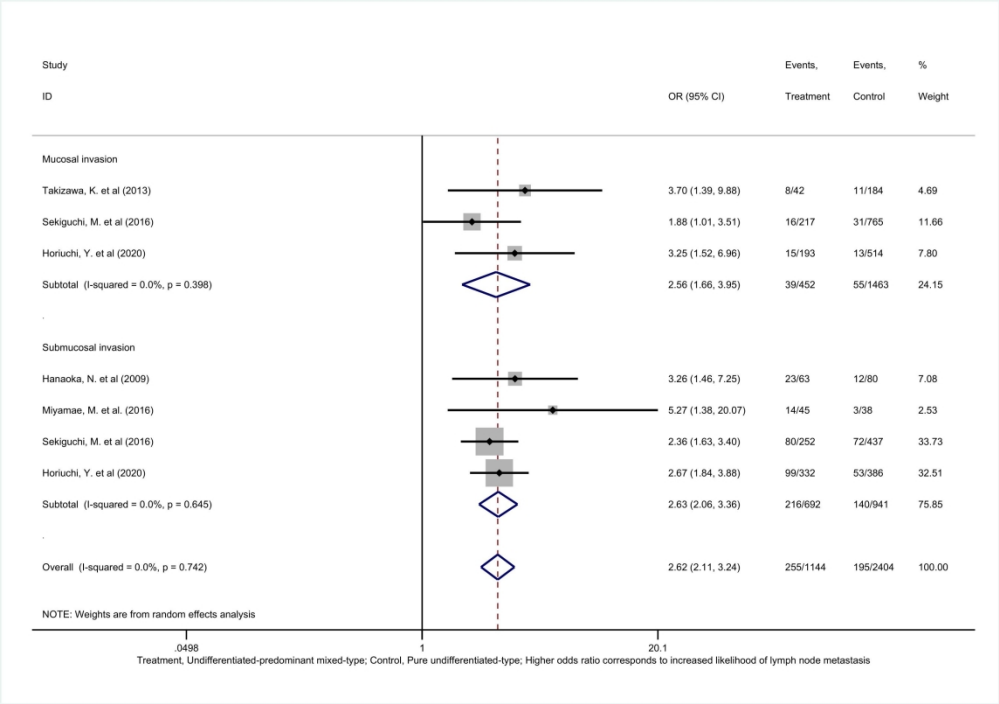
Flow chart summarizing the steps of identification for relevant studies.

338x190mm (300 x 300 DPI)



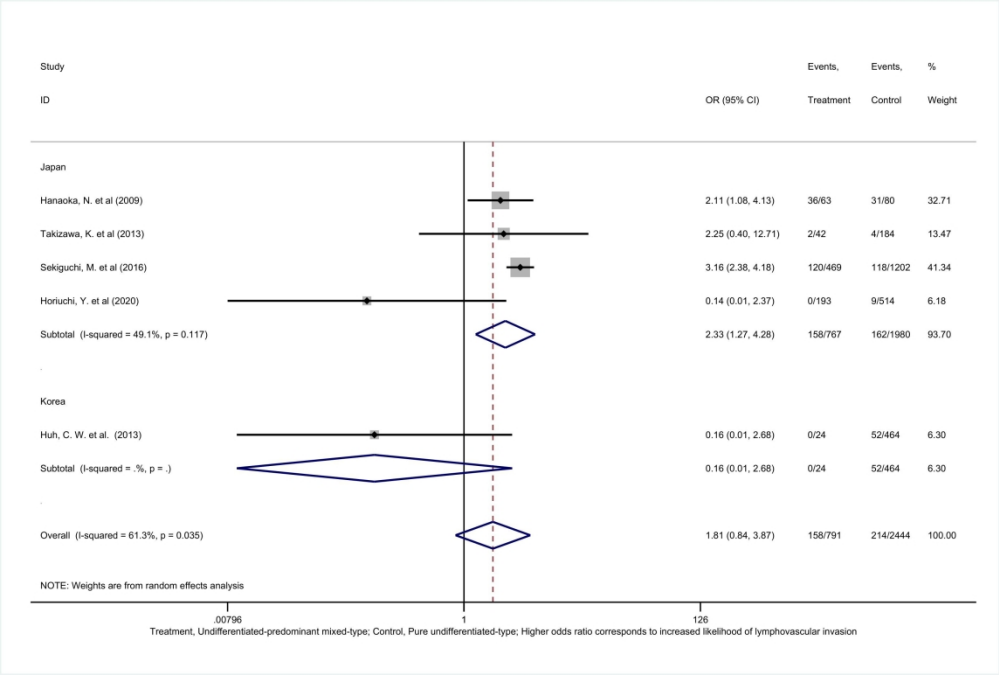
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis. Subgroup analysis was based on region.

276x208mm (300 x 300 DPI)



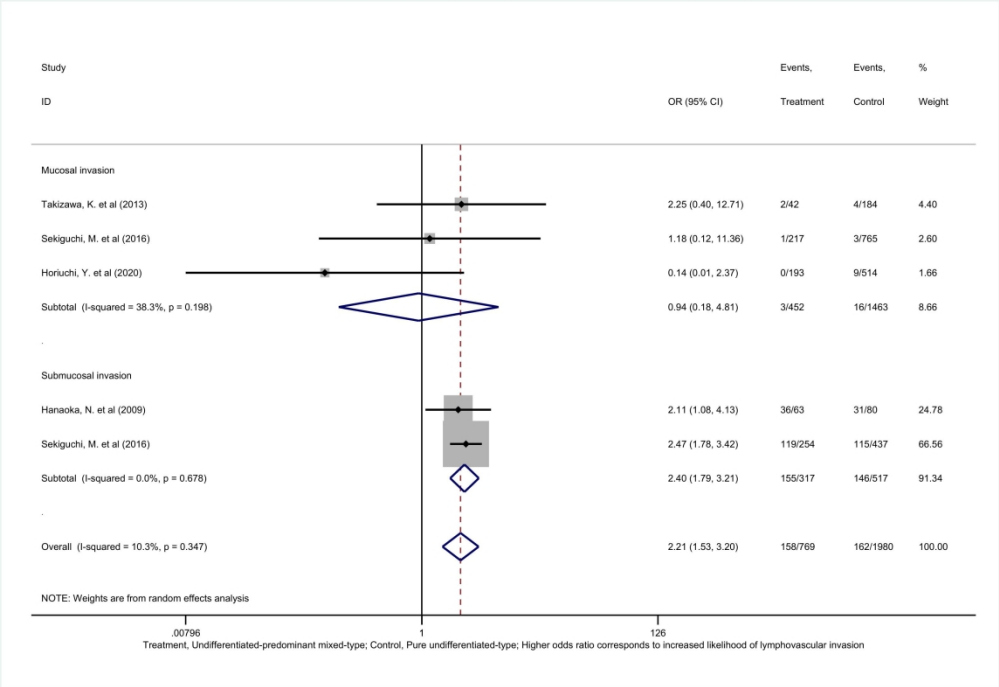
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis. Subgroup analysis was based on mucosal/submucosal invasion.

273x193mm (300 x 300 DPI)



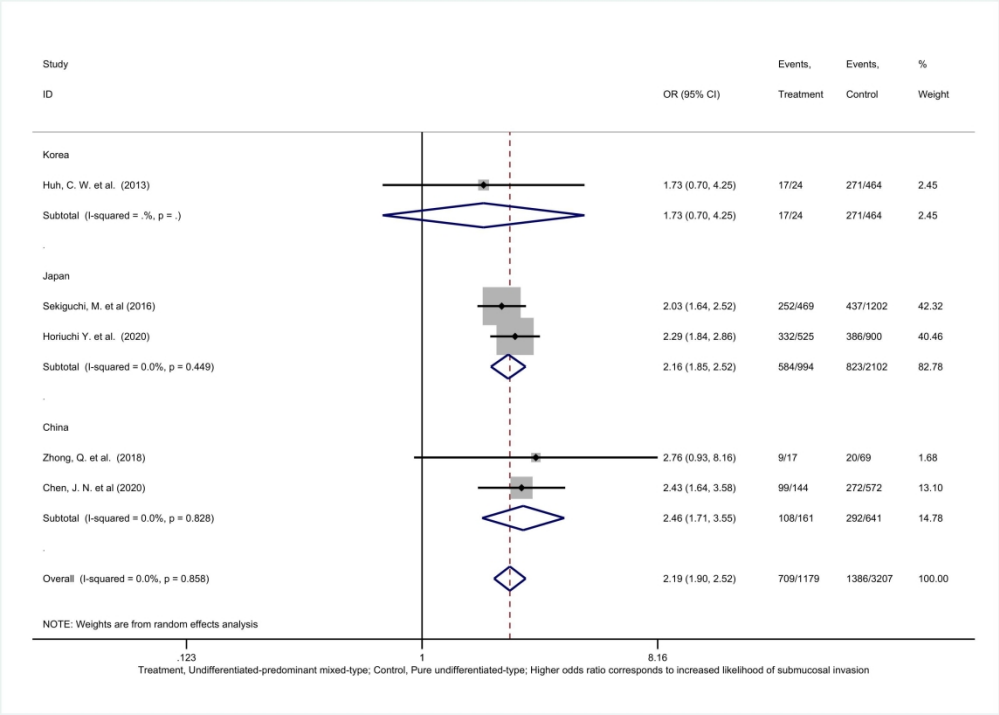
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of submucosal invasion. Subgroup analysis was based on region.

271x183mm (300 x 300 DPI)



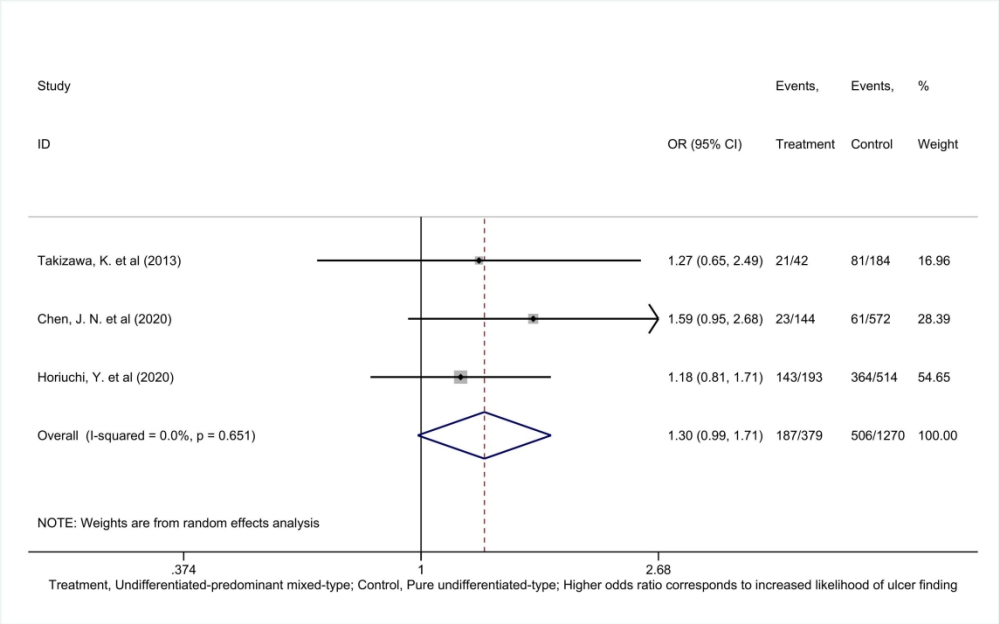
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymphovascular invasion. Subgroup analysis was based on region.

272x187mm (300 x 300 DPI)



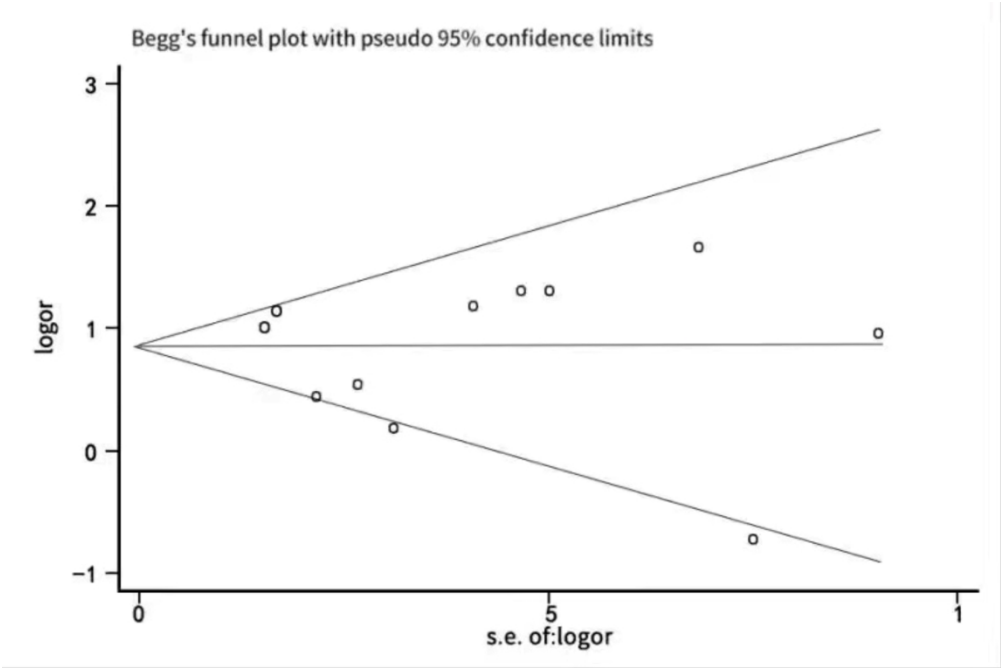
Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymphovascular invasion. Subgroup analysis was based on mucosal/submucosal invasion.

273x195mm (300 x 300 DPI)



Forest plots showing odds ratios (OR) with 95% confidence intervals (CI) comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of ulcer findings.

269x167mm (300 x 300 DPI)



Funnel plot comparing undifferentiated-predominant mixed-type with pure undifferentiated-type early gastric cancer (EGC) for the risk of lymph node metastasis.

282x188mm (300 x 300 DPI)

Search strategy in PubMed:

#1. (stomach neoplasma[MeSH Terms]) OR (early gastric cancer[Title/Abstract]) OR (EGC[Title/Abstract]) OR (early gastric carcinoma[Title/Abstract]) OR (mocosal[Title/Abstract]) OR (submocosal[Title/Abstract])

#2. (mixed type) OR (mixed histology) OR (undifferentiated) [Title/Abstract]

#3. #1 to #2

#4. #3 And English

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8-9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8-9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10-11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	13-14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	3-4

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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